



Machining Processes: Milling

Groover M.P.: Chapters 18 and 19



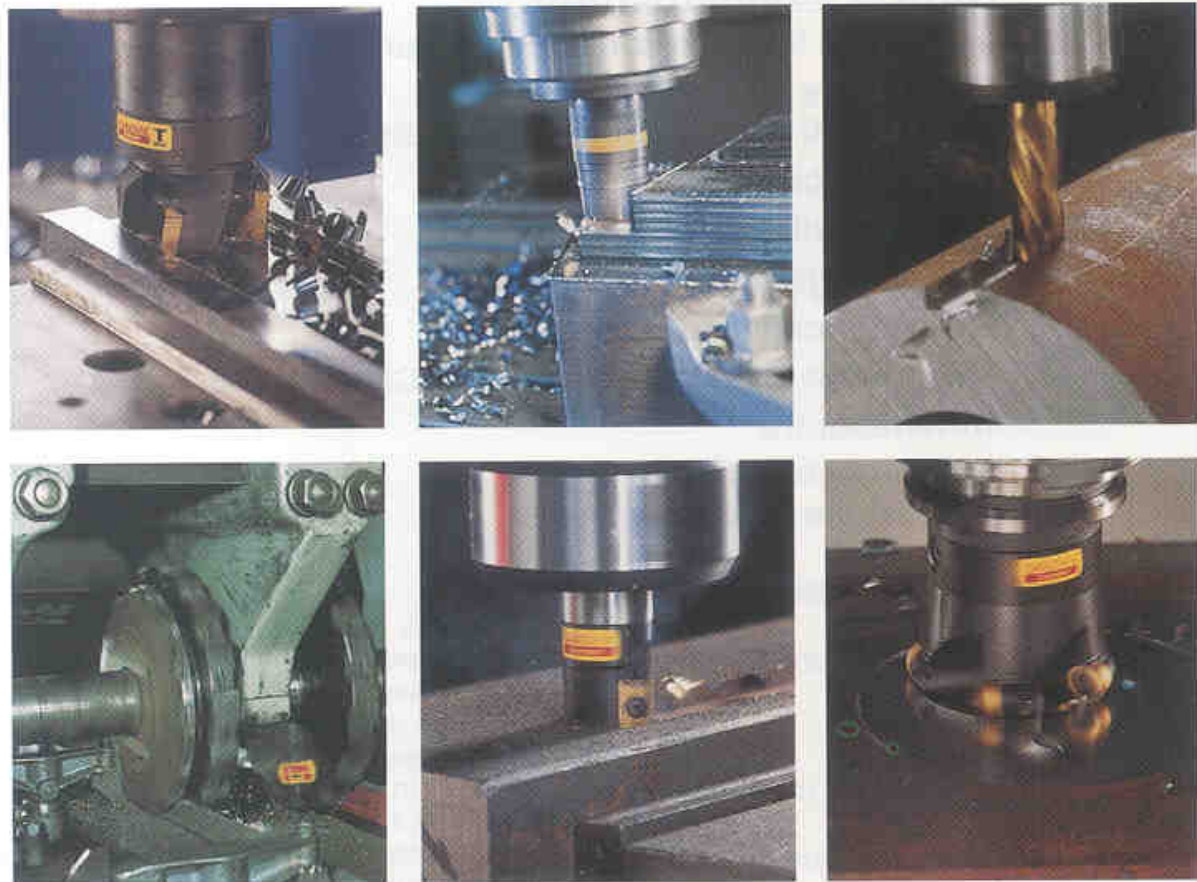
The student knows:

- basic principles, tools and machine tools;
- how to evaluate cutting force, power and work;
- how to define the appropriate process parameters.





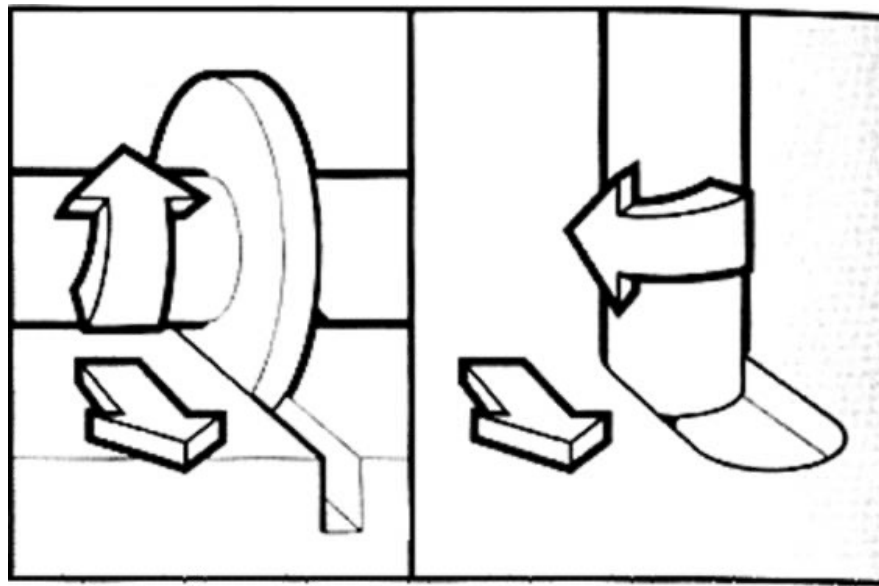
Goal: to obtain a wide range of surfaces (planes, slots, shoulders, etc.)





The tool (mill) is circular with a set of cutting edges that are cyclically cutting.

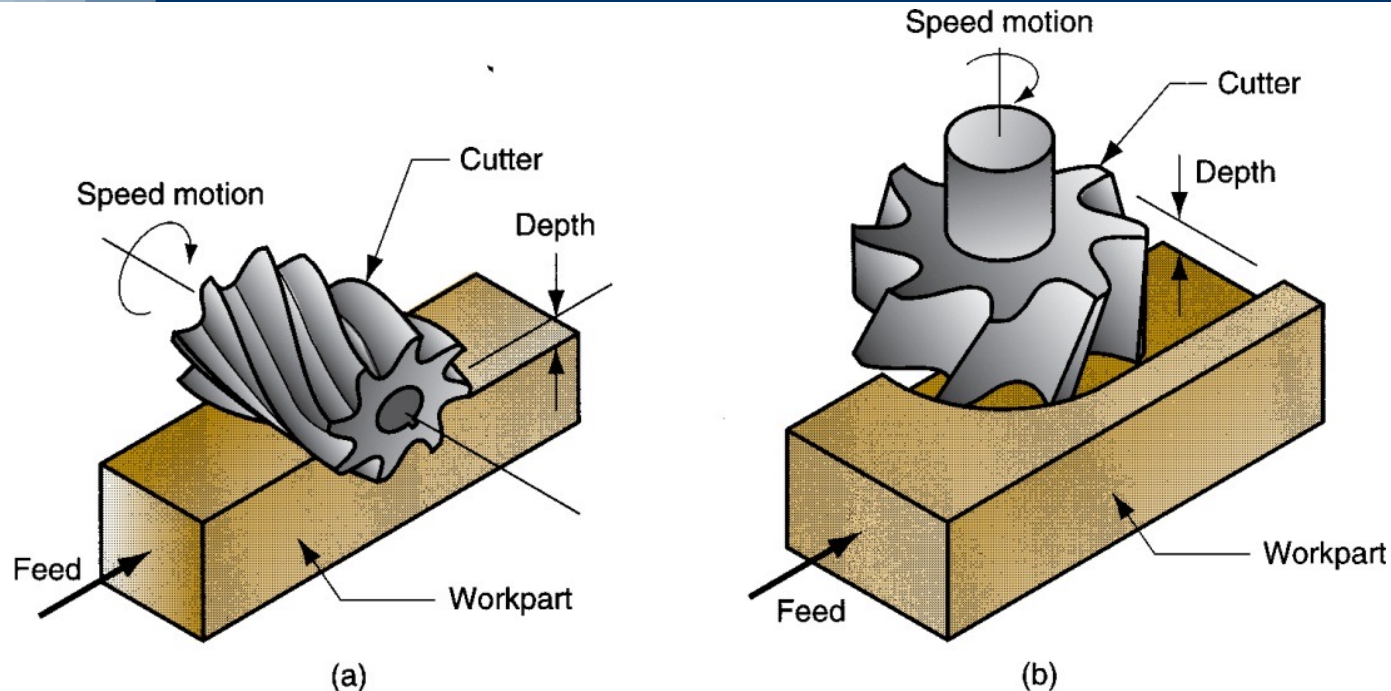
- The cutting motion (v_c) is held by the tool;
- The feed motion (v_f) is held by the part and/or by the tool.





TWO FORMS OF MILLING

5



Peripheral milling (a)

- Cutter axis parallel to surface being machined
- Cutting edges on outside periphery of cutter

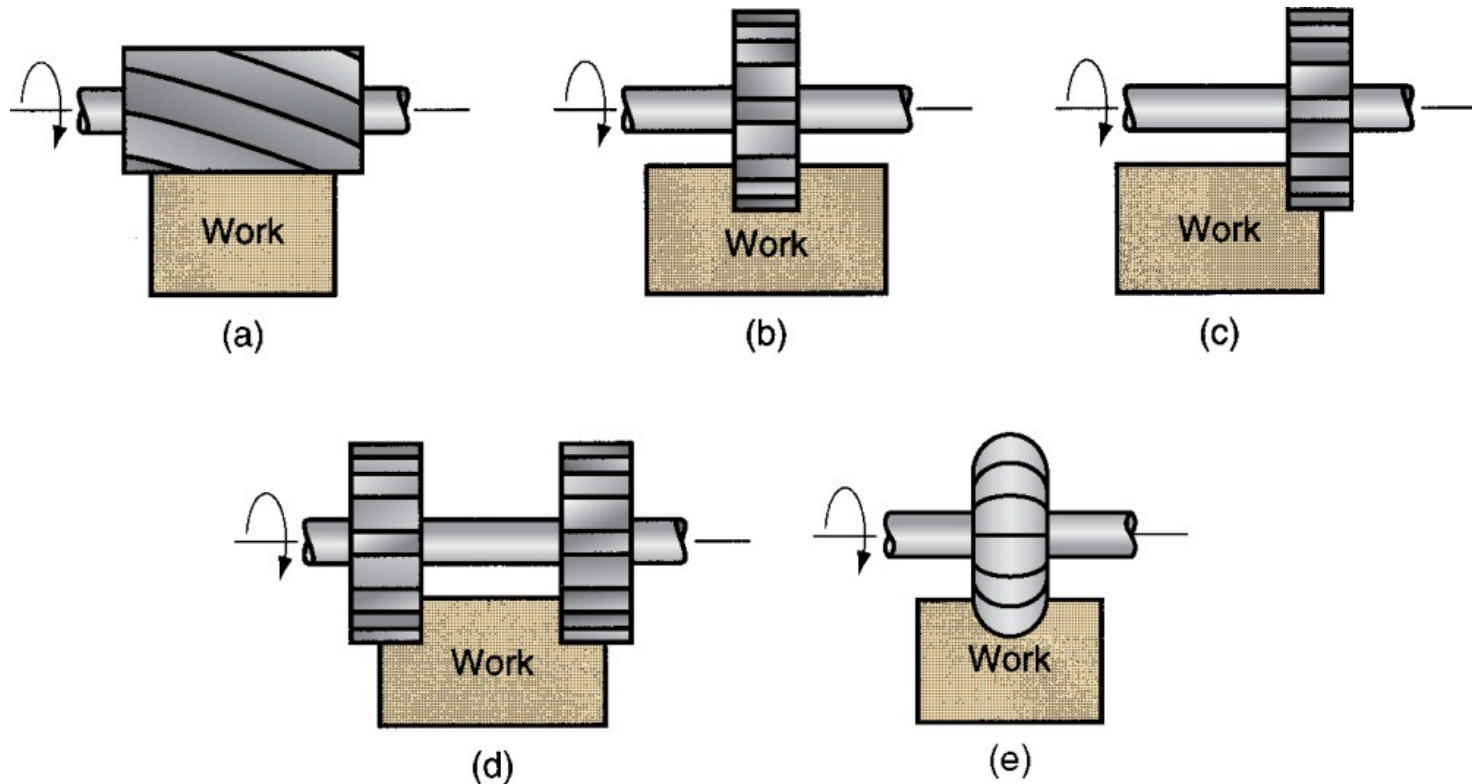
Face milling (b)

- Cutter axis perpendicular to surface being milled
- Cutting edges on both the end and outside periphery of the cutter



TYPES OF PERIPHERAL MILLING

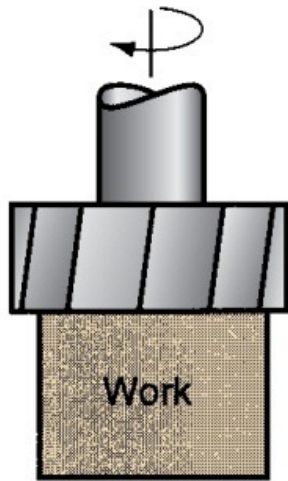
(a) Slab milling, (b) Slotting, (c) Side milling,
(d) Straddle milling, and (e) Form milling



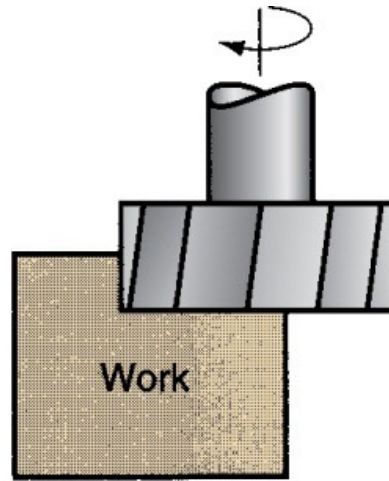


TYPES OF FACE MILLING

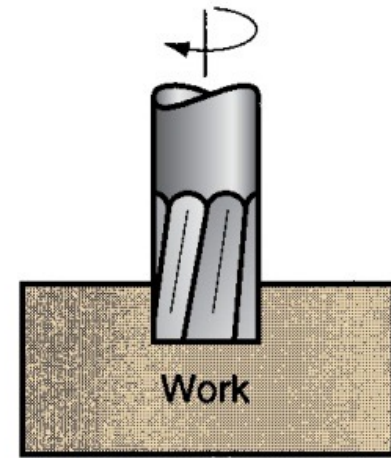
(a) Conventional face milling, (b) Partial face milling, and (c) End milling.



(a)



(b)

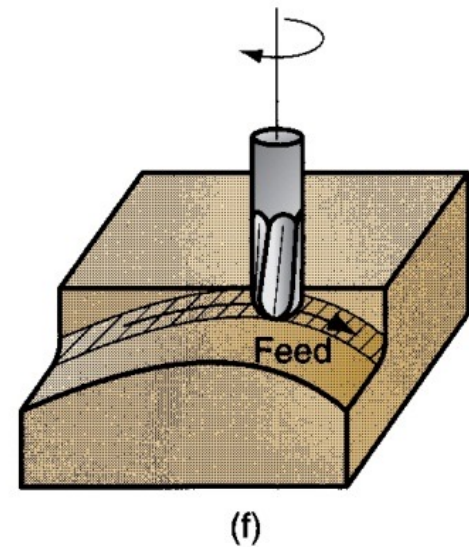
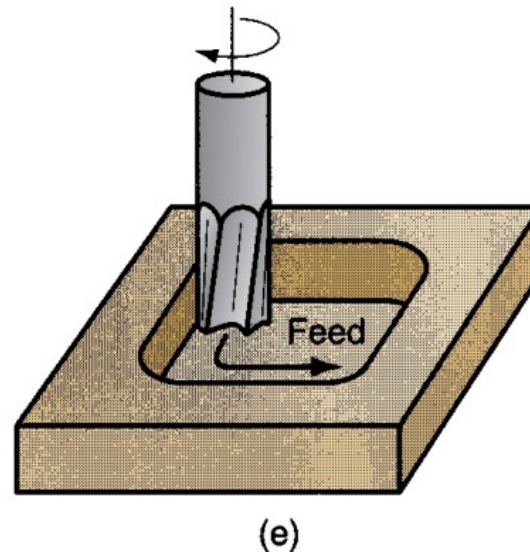
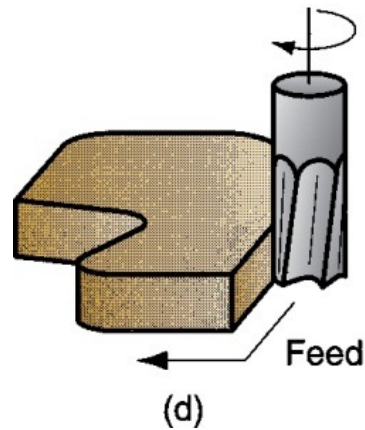


(c)



TYPES OF FACE MILLING

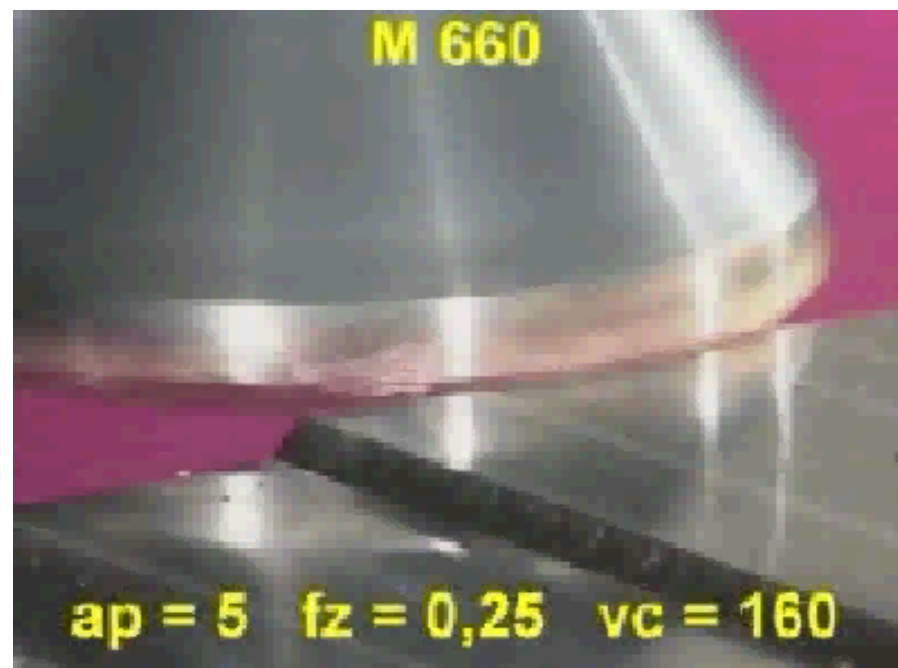
(d) Profile milling, (e) Pocket milling,
and (f) Surface contouring





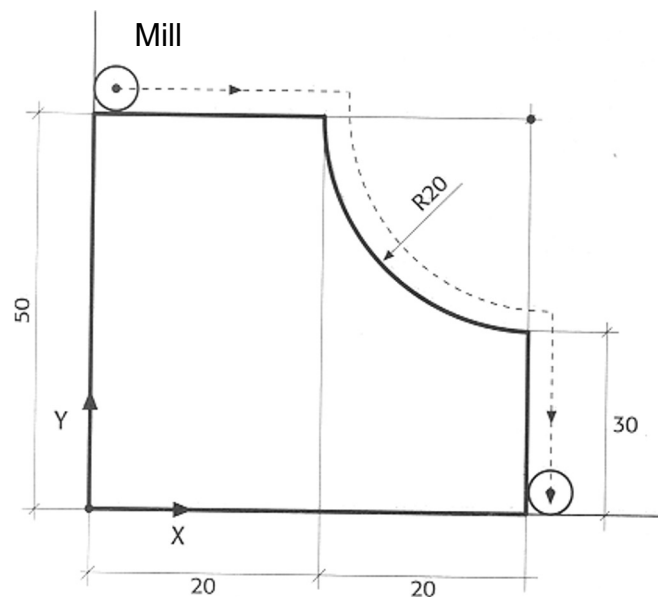
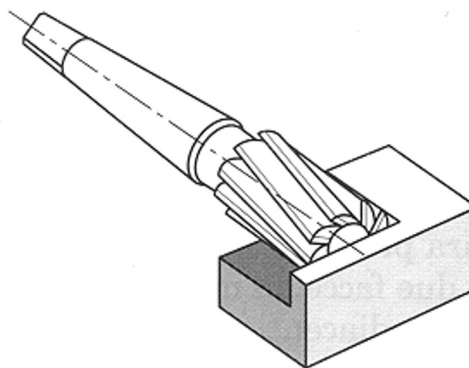
FACE MILLING

9



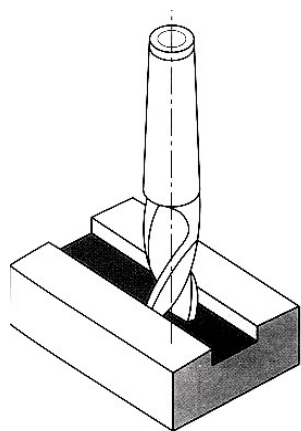


Contour/End milling is the machining of profiles with end mills.

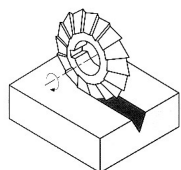
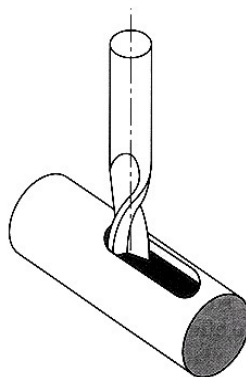




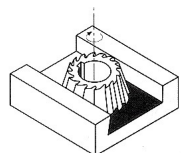
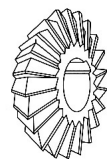
Slotting is the operation by which slots of different geometries and sizes are obtained.



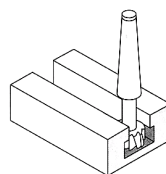
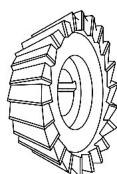
c)



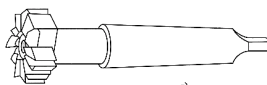
a)



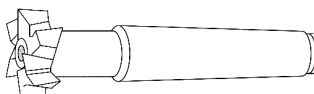
b)



a)



b)



Other types of slots: machining of T, V, or U-shaped slots.



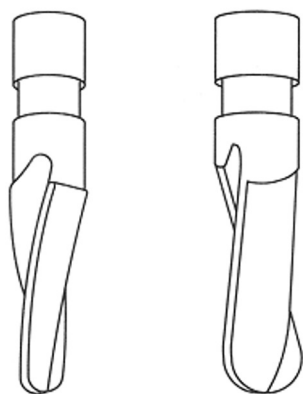
END MILLING AND SLOTTING

12





For example, the machining of dies and molds.

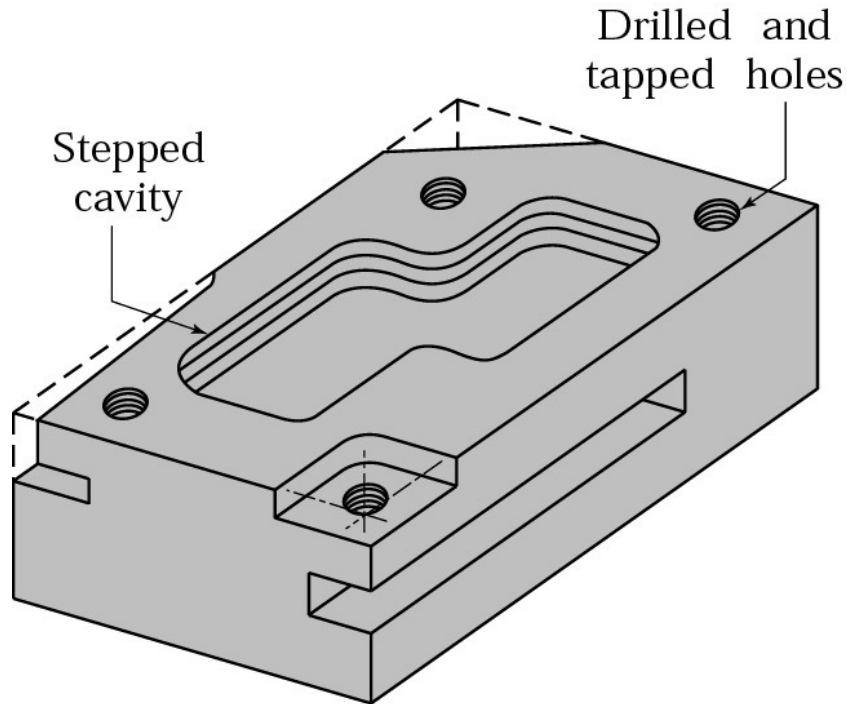


d)





TYPICAL MACHINED PARTS



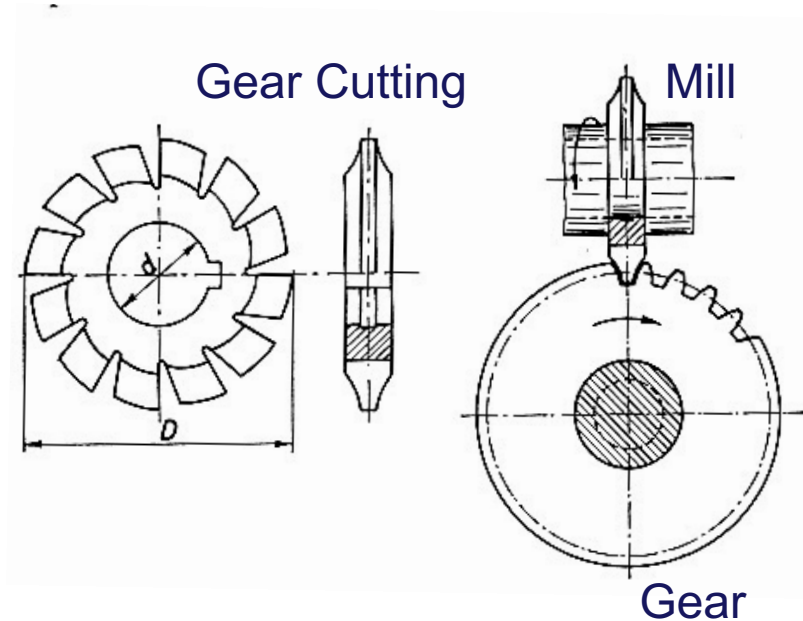
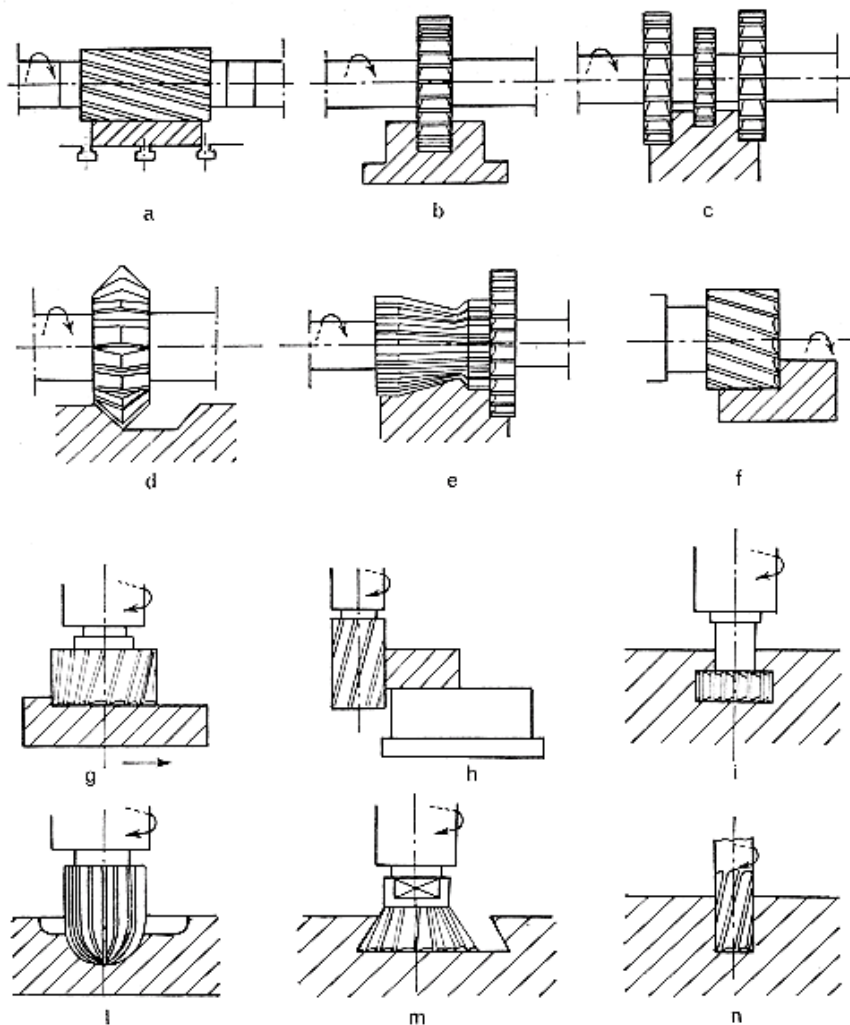
A typical part that can be produced on a milling machine equipped with computer controls.

Such parts can be made efficiently and repetitively on computer numerical control (CNC) machines, without the need for re-fixturing or re-clamping the workpiece.



EXAMPLES OF MILLING OPERATIONS

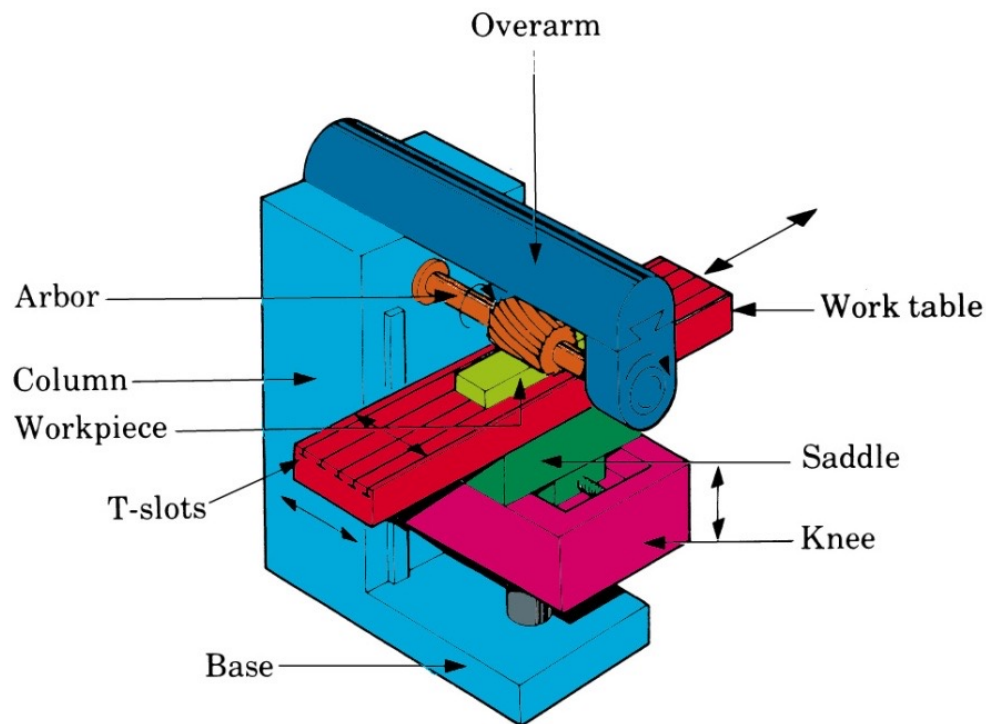
There exist a large variety of mills for different geometries





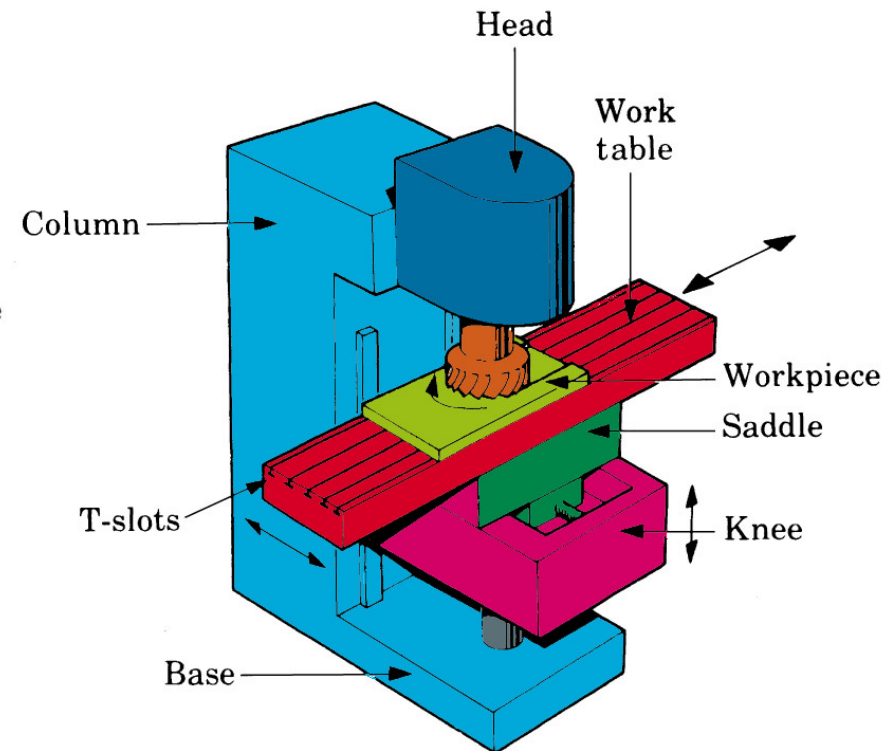
Summarizing, milling is a machining operation in which work is fed past a rotating tool with multiple cutting edges:

- Axis of tool rotation is perpendicular to feed
- Cutting tool called a milling cutter
 - Cutting edges called teeth
- Machine tool called a milling machine
- Interrupted cutting operation
- Basic milling operation creates a planar surface
 - Other geometries possible



Schematic illustration of a horizontal-spindle column-and-knee type milling machine.

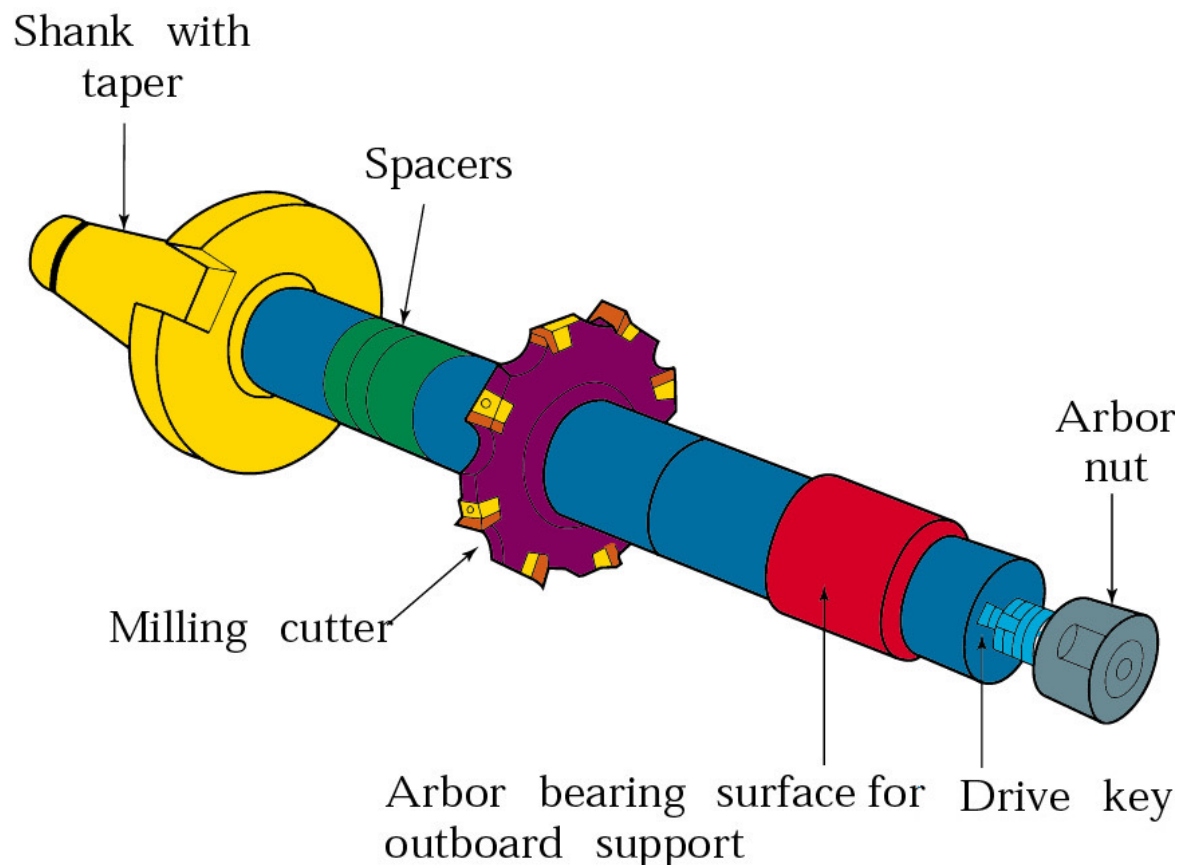
Source: G. Boothroyd.



Schematic illustration of a vertical-spindle column-and-knee type milling machine (also called a *knee miller*).

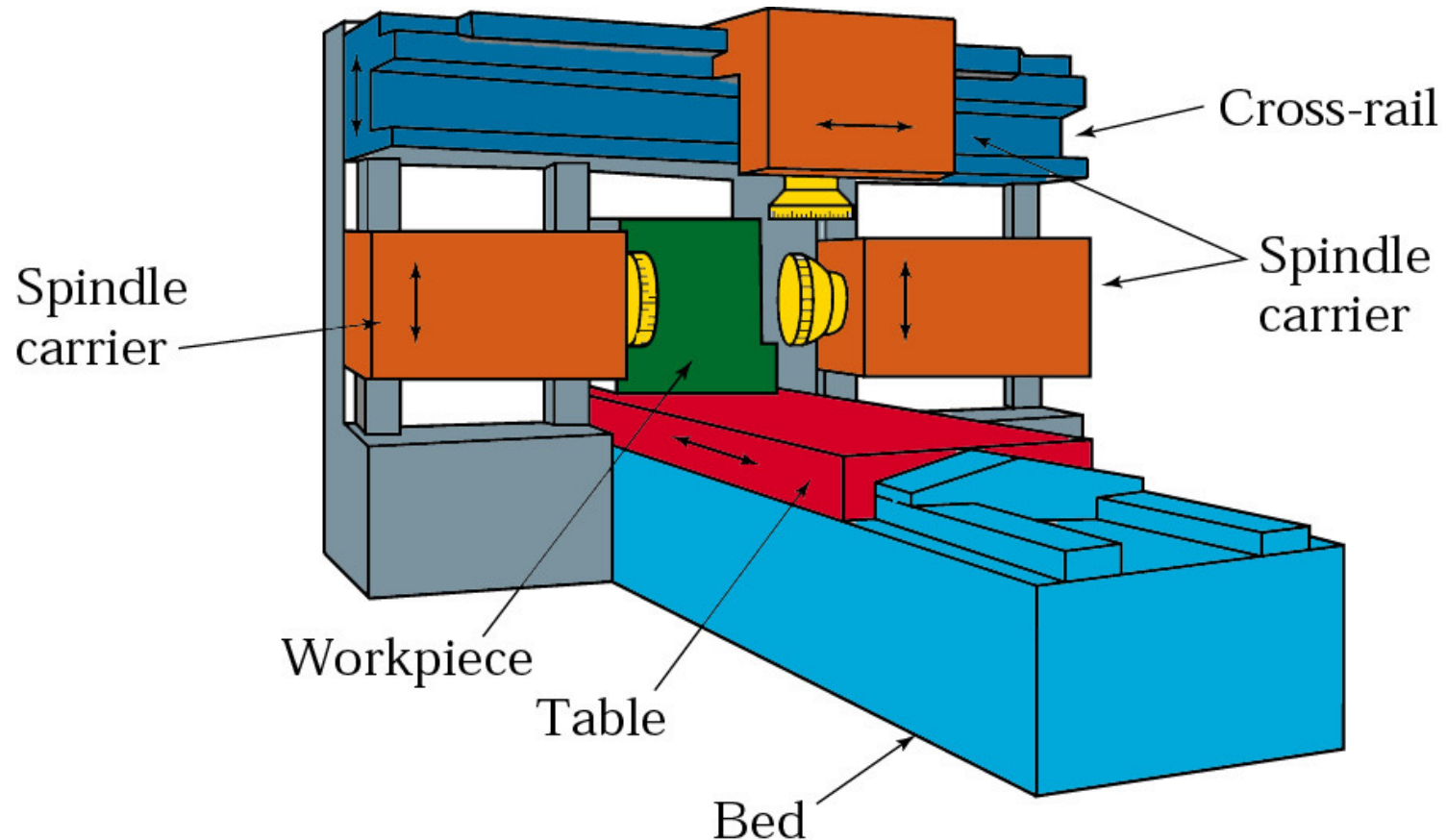
Source: G. Boothroyd.

Mounting a milling cutter on an arbor for use on a horizontal milling machine.





BED-TYPE MILLING MACHINE

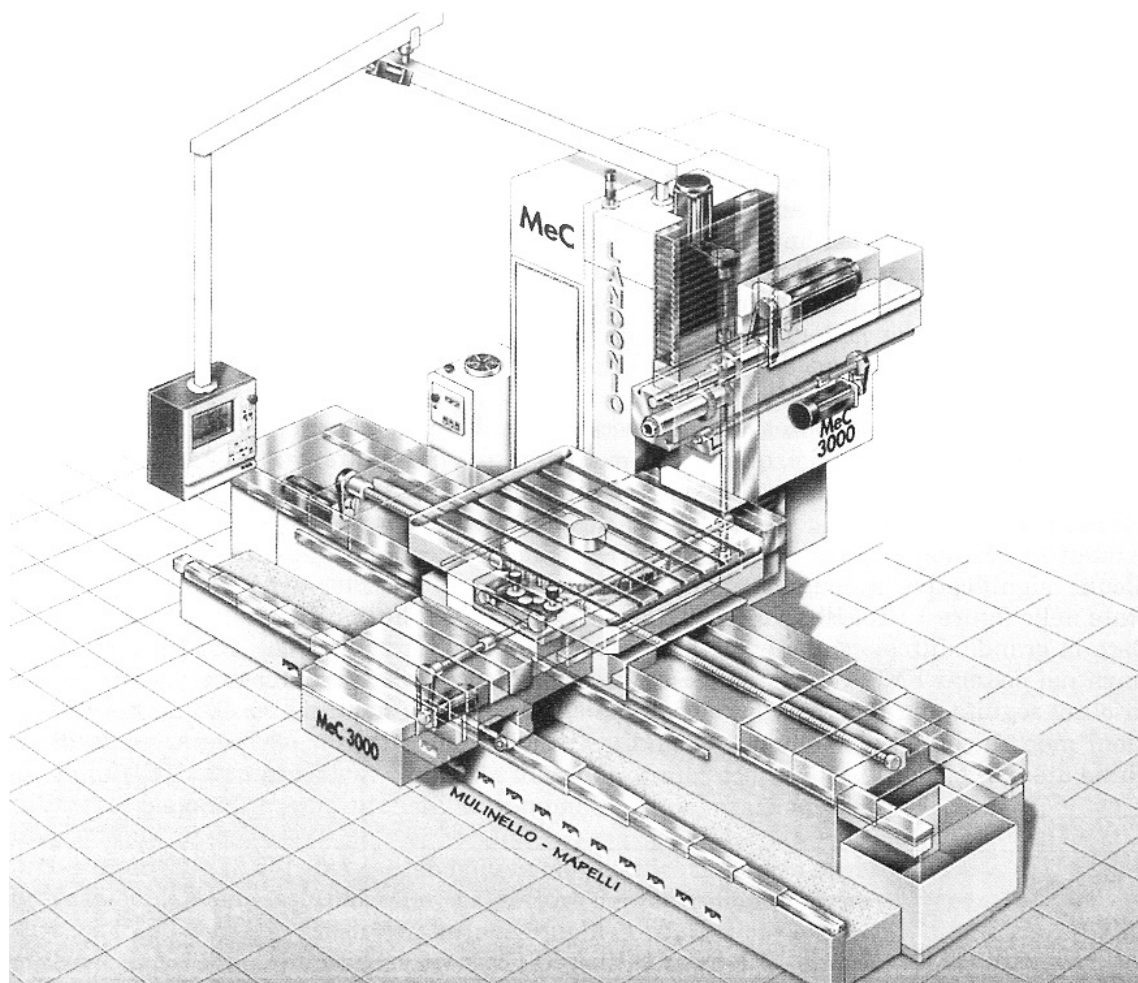


Schematic illustration of a bed-type milling machine. Note the single vertical-spindle cutter and two horizontal spindle cutters.

Source: ASM International.



Modern CNC milling machine





<https://youtu.be/UUAFu-TeQAw>

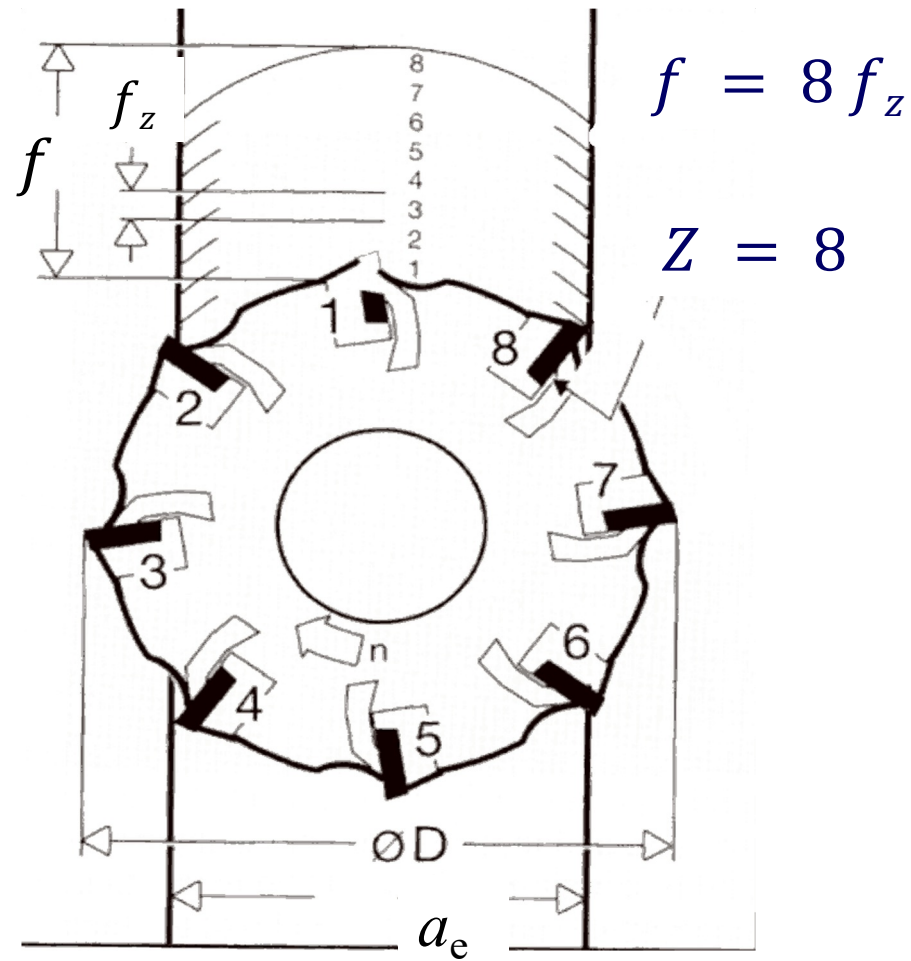


Characteristic machining parameters:

- Number of cutters (teeth) of the mill: Z
- Feed per tooth: f_z [mm/(rev*tooth)]
- Mill diameter: D [mm]
- Rotational speed of the mill: n [rpm]
- Feed velocity: $v_f = f_z Z n$ [mm/min]
- Cutting speed: $v_c = \pi D n / 1000$ [m/min]

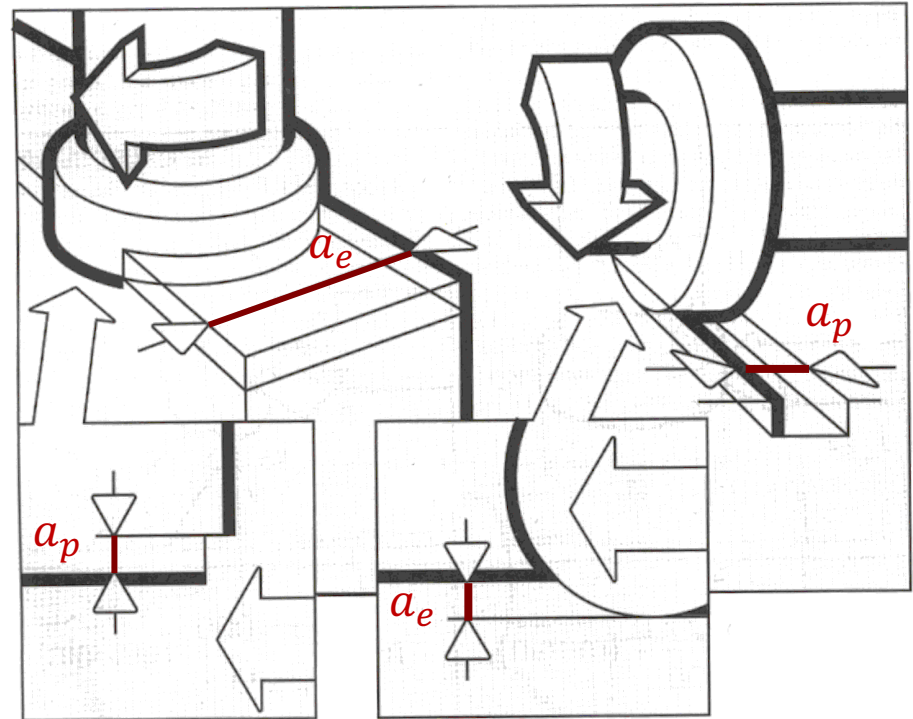
Typical values:

- Mill diameter $D = 0,2 \div 2000$ [mm]
- Number of teeth of the mill $Z = 2 \div 200$ [teeth]
- Feed per tooth $f_z = 0,01 \div 1$ [mm/(tooth-rev)]
- Cutting speed $v_c = 30 \div 500$ [m/min]



Axial depth of cut a_p in face milling or radial cutting depth a_e in slab milling is the thickness of the material to be removed.

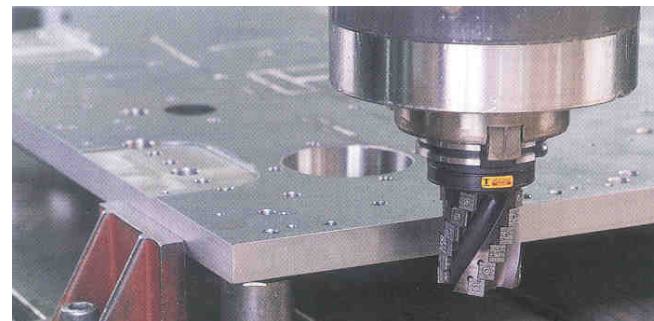
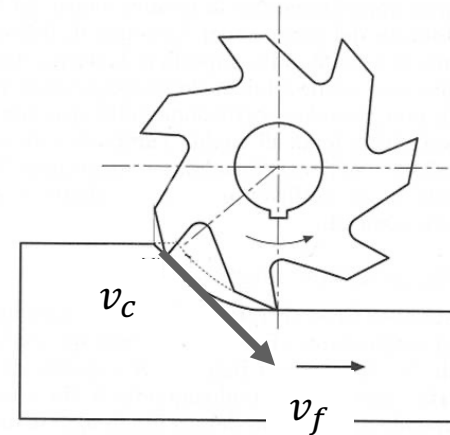
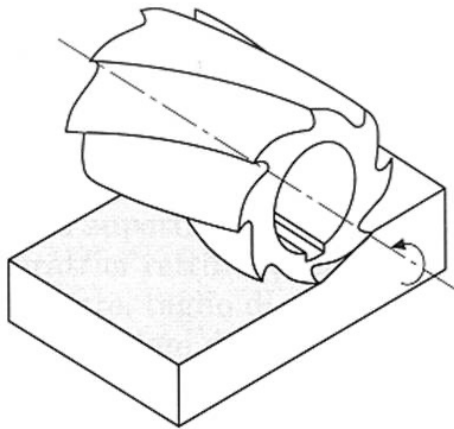
Radial depth of cut a_e in face milling or axial depth of cut a_p in slab milling is the width of the material to be removed.



PERIPHERAL MILLING: SLAB MILLING

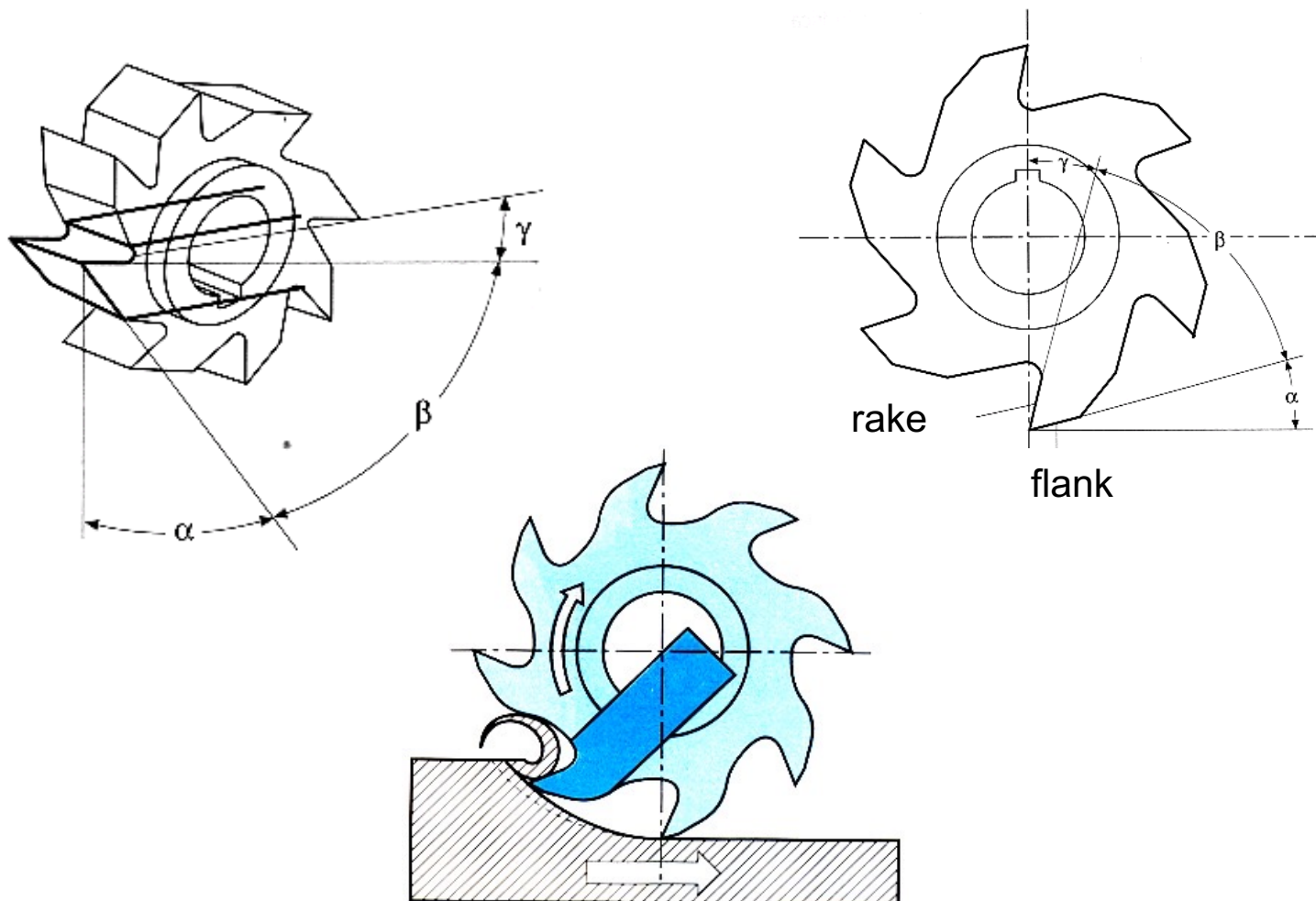
The tool axis is parallel to the machined surface.

The part of the mill interested by the cutting operation is the cylindrical part.





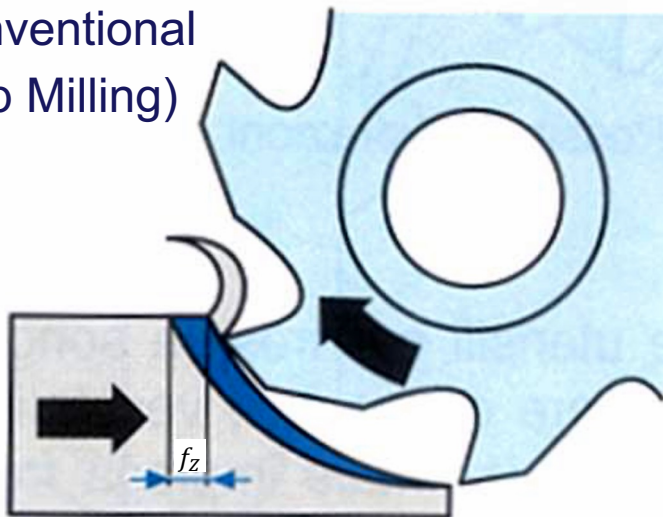
The cutter: peripheral mill with straight cutting edges.



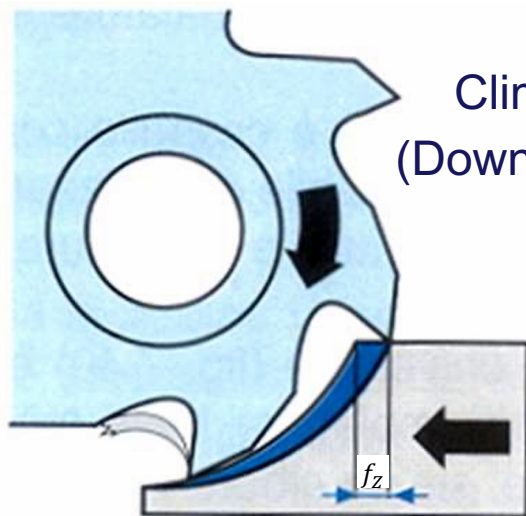


FEED MOTIONS IN SLAB MILLING

Conventional
(Up Milling)



Climbing
(Down Milling)



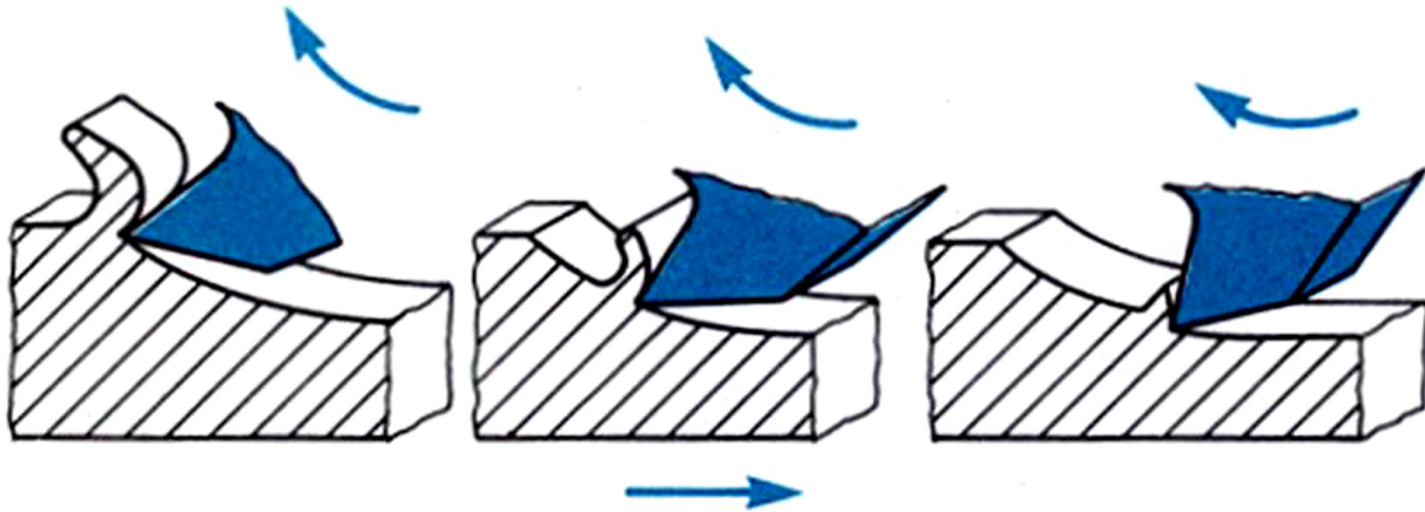
SANDVIK
Coromant



CONVENTIONAL OR UP MILLING

The chip thickness increases during the cutting.

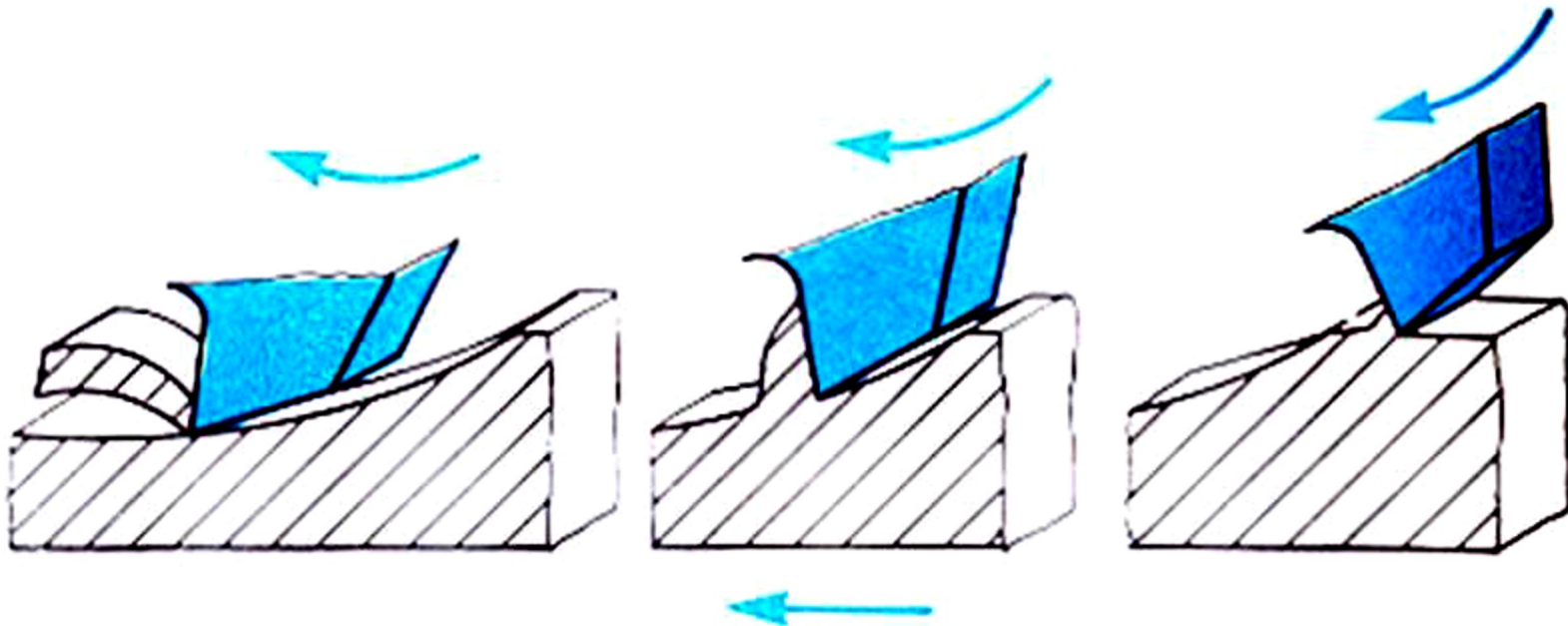
The cutter slides before to start cutting: tool wear, heating.

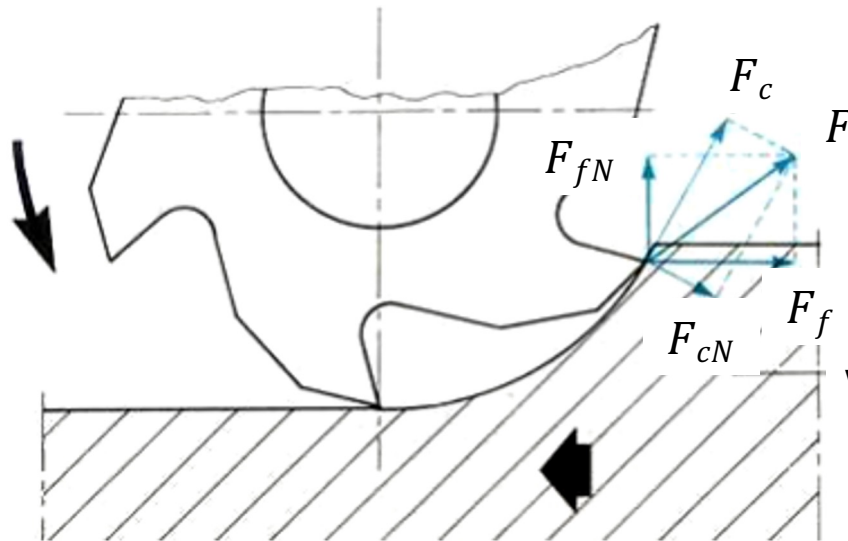




CLIMBING OR DOWN MILLING

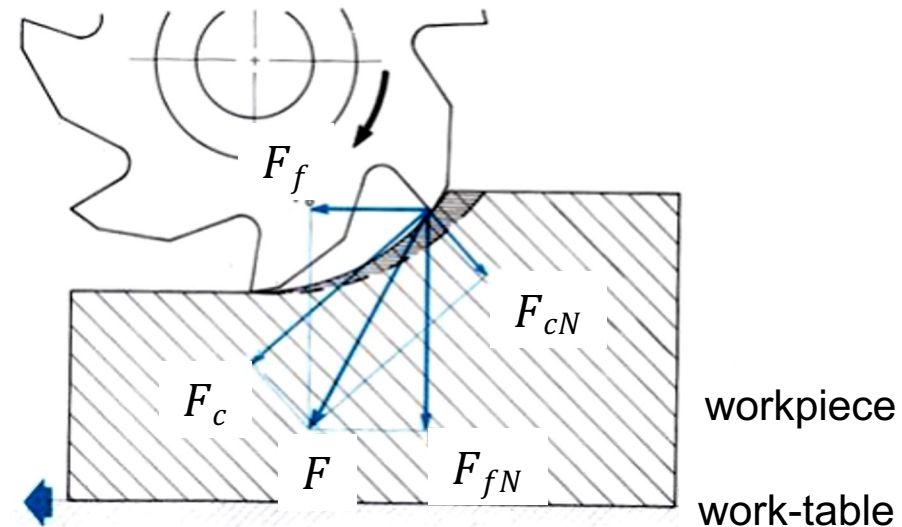
The chip thickness decreases during the cutting.
The tool starts to cut with large chip thicknesses:
the initial shock may cause the tool fracture.

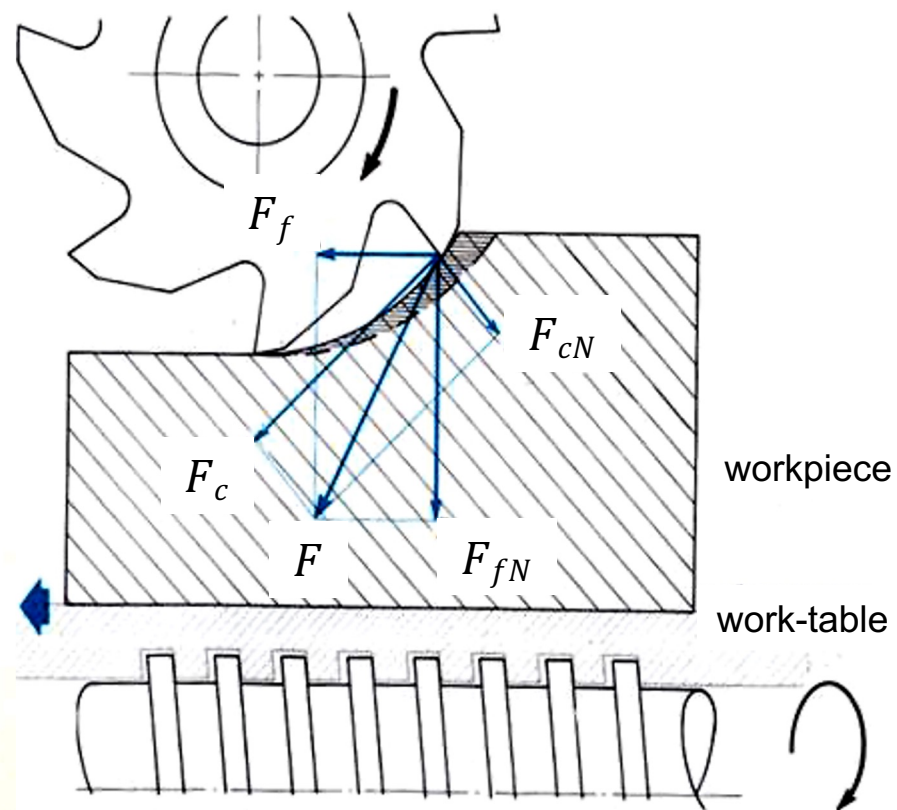
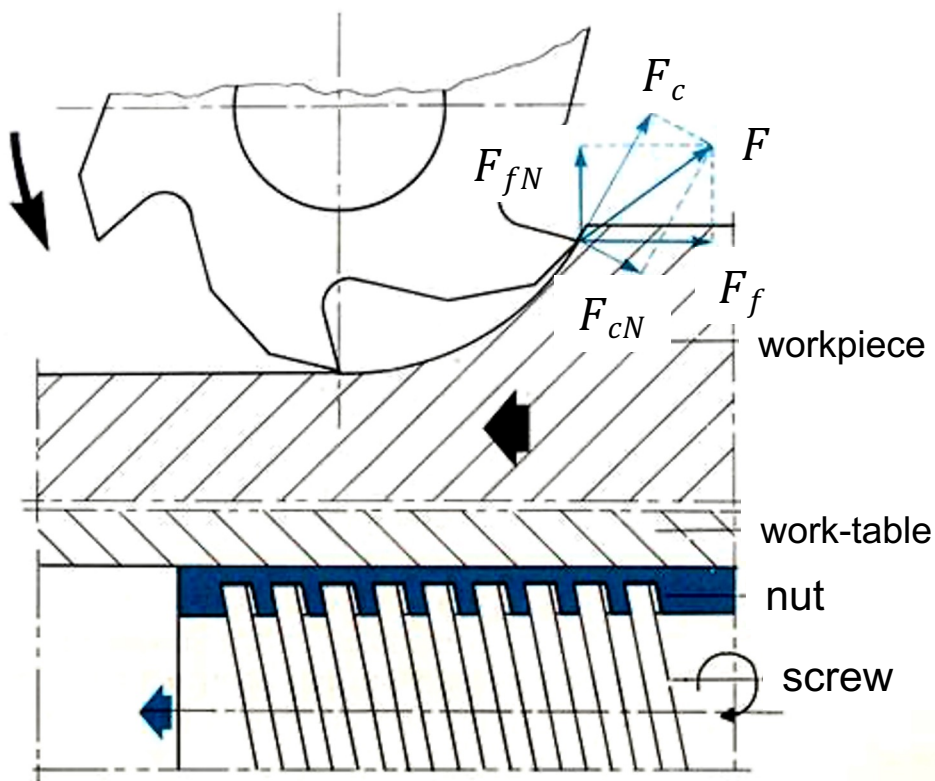




- Increased load on the work-holding device
→ vibration

- Reduced load on the work-holding device
- Possible problem in motion transmission
→ vibration

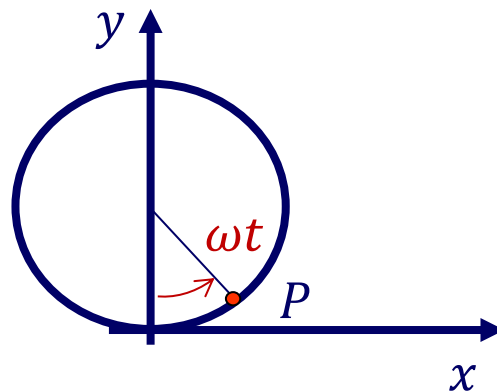






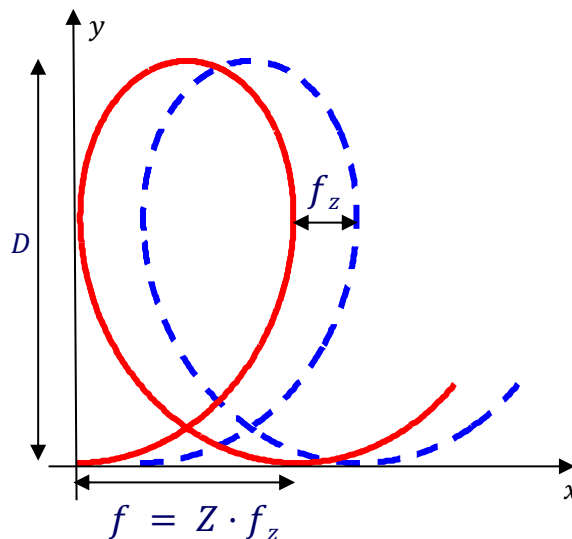
Combination of two motions: rotational and linear.

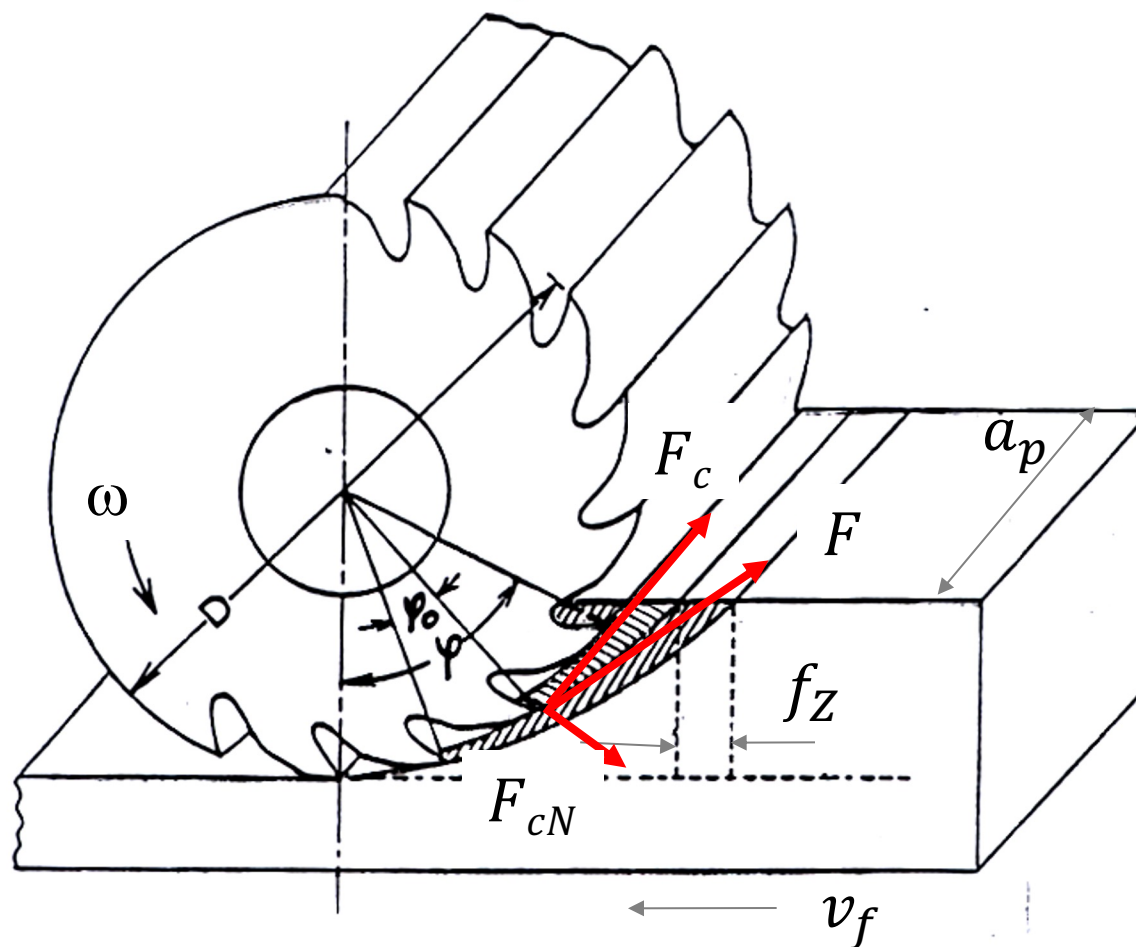
P: position of
the tooth



$$\begin{cases} y = \frac{D}{2} (1 - \cos \omega t) \\ x = \frac{D}{2} \sin \omega t + v_f t \end{cases}$$

Tooth
trajectory:
cycloid



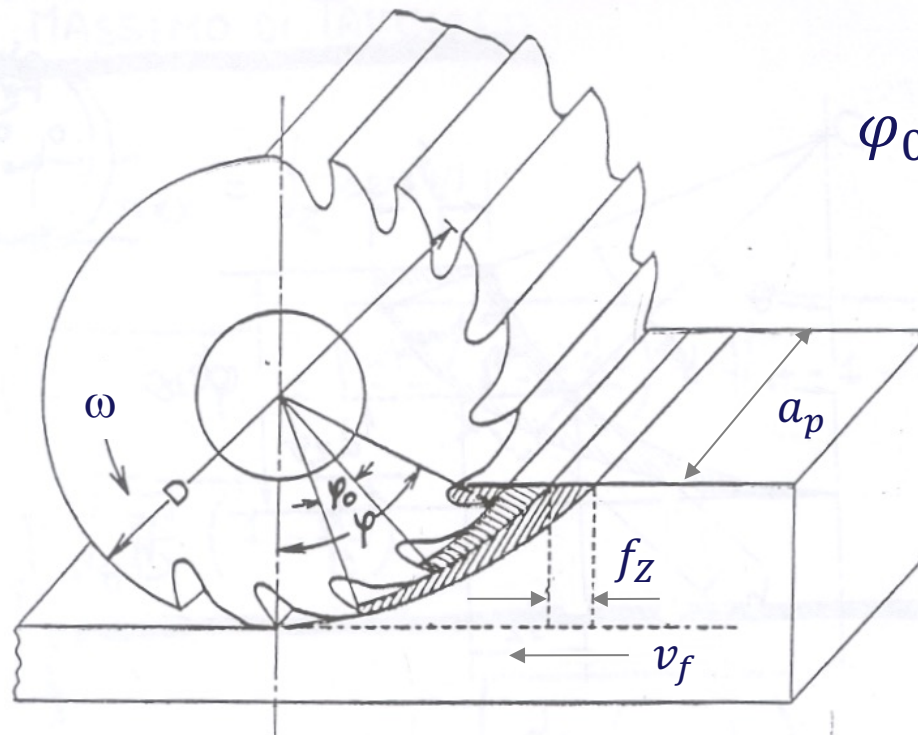




Number of working teeth z

$$z = \frac{\varphi}{\varphi_0}$$

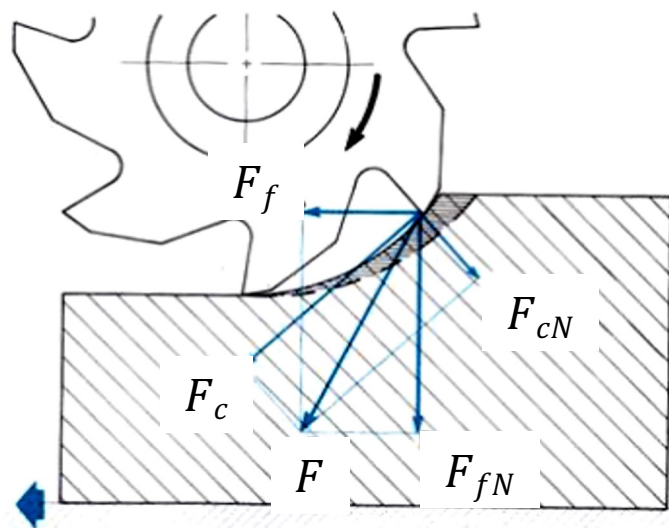
$$\varphi_0 = \frac{2\pi}{Z}$$





CUTTING POWER IN SLAB MILLING

Decomposition of the forces in slab milling:



$$F_c(\theta) = k_{c,\theta} A_{D,\theta} \text{ [N]}$$

$$\text{Where } k_{c,\theta} = k_{cs} h_{D,\theta}^{-x}$$

Therefore each tooth gives the following contribution to the torque:

$$T_{c,\theta} = F_c(\theta) \frac{D/2}{1000} = k_{c,\theta} A_{D,\theta} \frac{D/2}{1000} \text{ [Nm]}$$

The total torque is:

$$T_c = \sum_{i=1}^z F_c(\theta_i) \frac{D/2}{1000} \text{ [Nm]}$$

It follows, the cutting power is: $P_c = T_c \omega$ [W]

$$\text{Where } \omega = \frac{2\pi n}{60} = \frac{1000 \cdot v_c}{D/2} \left[\frac{\text{rad}}{\text{s}} \right]$$



CUTTING POWER IN SLAB MILLING

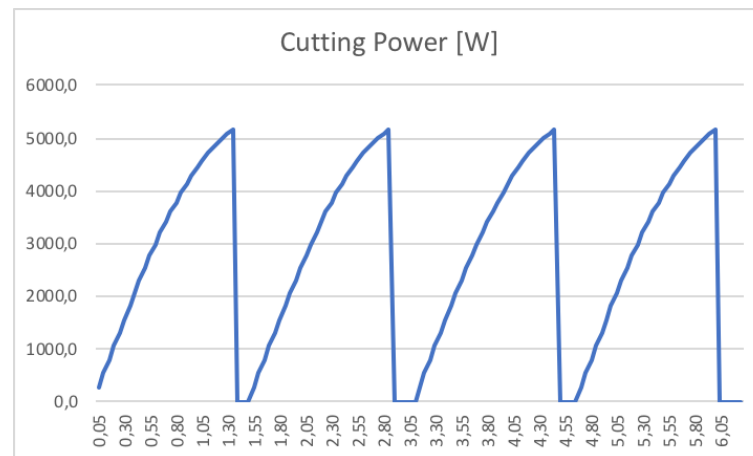
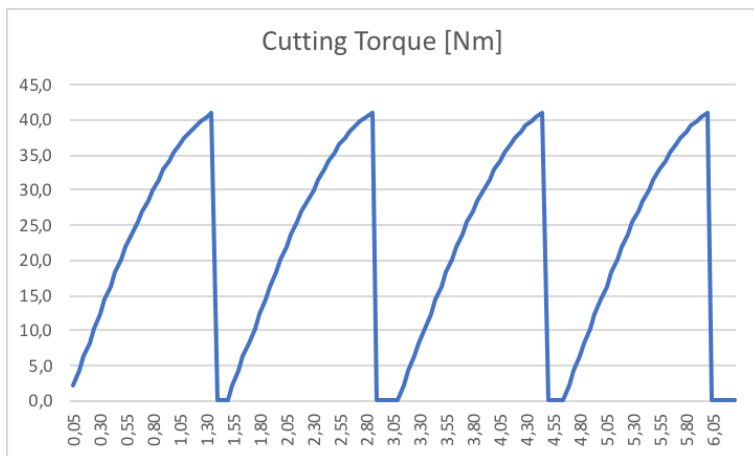
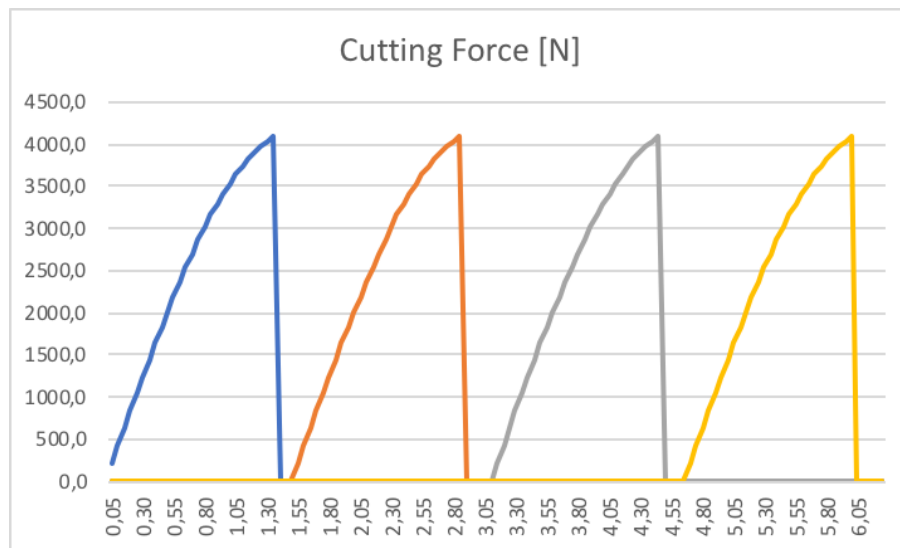
Case 1:

$Z = 4, z = 0,84$

$D = 20 \text{ mm}$

$a_e = 7,5 \text{ mm}$

$a_p = 20 \text{ mm}$



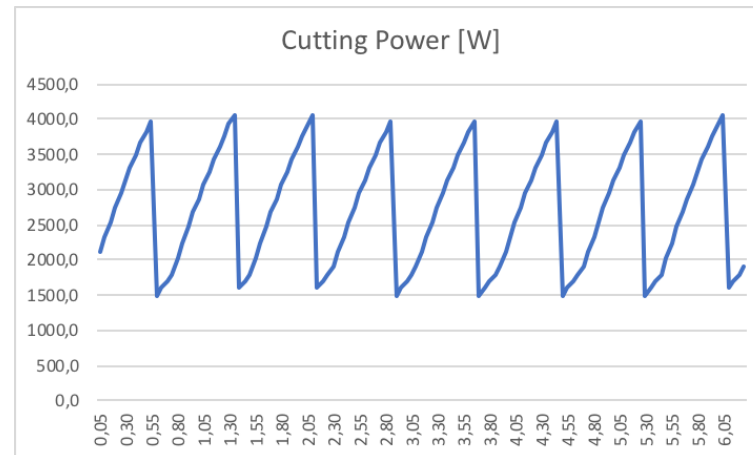
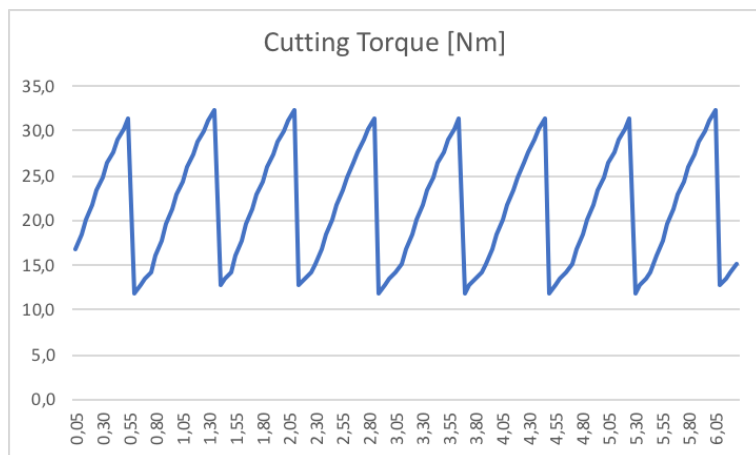
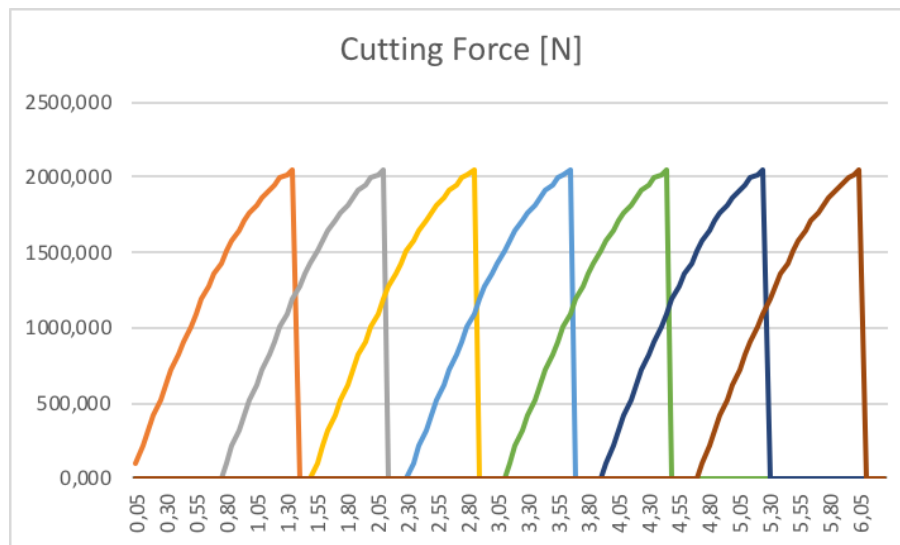


CUTTING POWER IN SLAB MILLING

38

Case 2:

$Z = 8, z = 1,68$

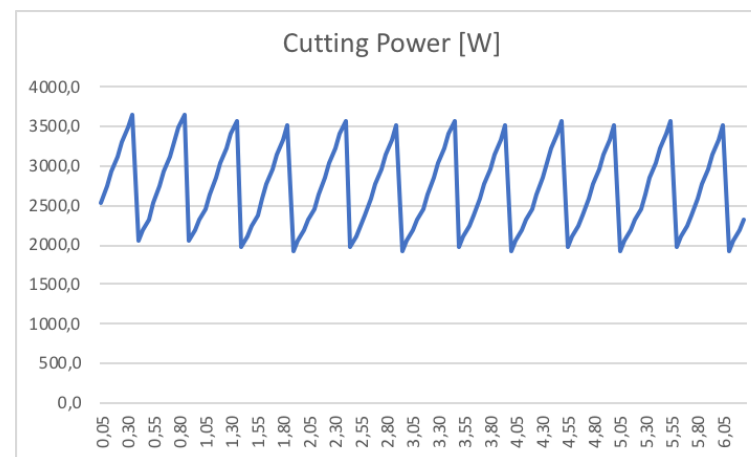
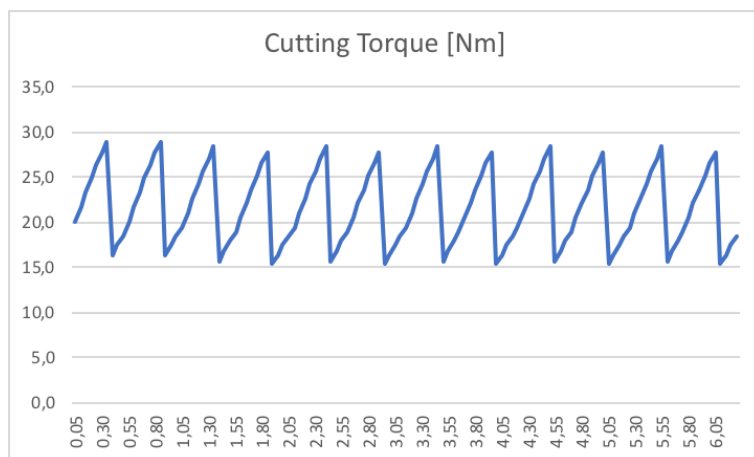
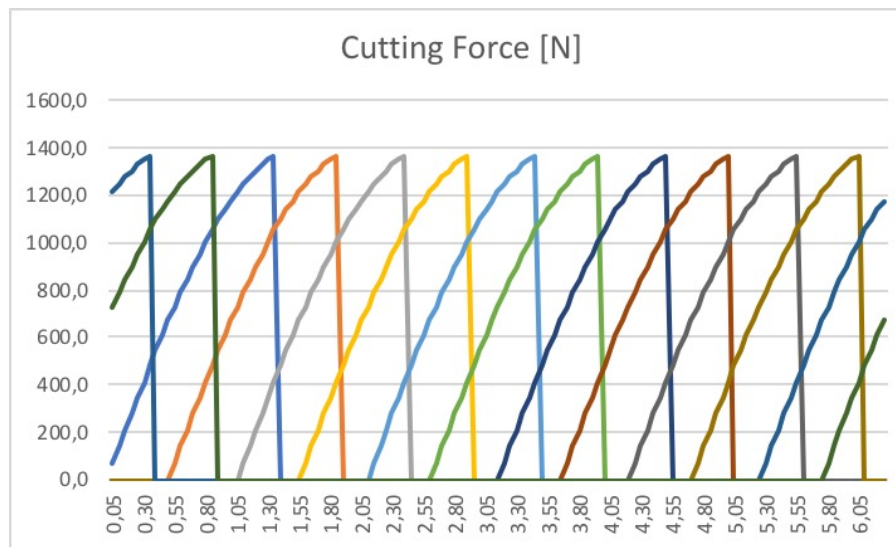




CUTTING POWER IN SLAB MILLING

Case 3:

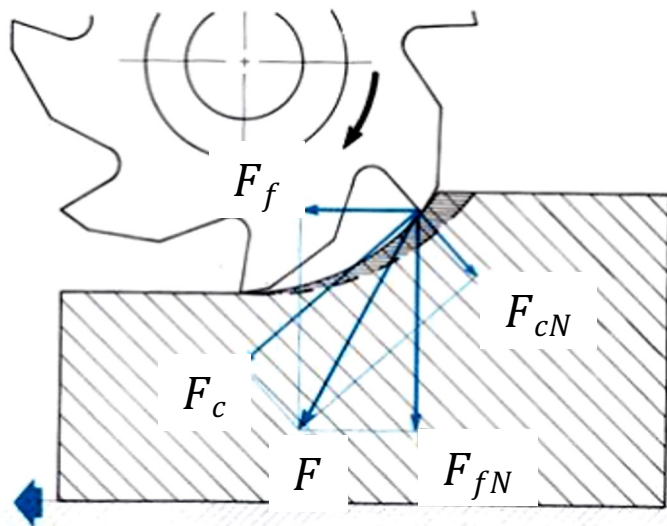
$Z = 12, z = 2,52$





CUTTING POWER IN SLAB MILLING

If $z \leq 1$, then it applies the worst case approach



$$h_{D,max} = h_D(\theta = \varphi) = f_Z \sin \varphi$$

$$A_{D,max} = A_D(\theta = \varphi) = f_Z \sin \varphi \cdot a_p$$

$$F_{c,max} = k_{c,max} A_{D,max} \text{ [N]}$$

$$\text{Where } k_{c,max} = k_{cs} h_{D,max}^{-x}$$

Therefore the torque is:

$$T_c = F_{c,max} \frac{D/2}{1000} = k_{c,max} A_{D,max} \frac{D/2}{1000} \text{ [Nm]}$$

$$\text{It follows: } P_c = T_c \omega \text{ [W]}$$

$$\text{Where } \omega = \frac{2\pi n}{60} = \frac{1000 \cdot v_c}{D/2} \left[\frac{\text{rad}}{\text{s}} \right]$$



CUTTING POWER IN SLAB MILLING

If $z > 2$, then it applies the average h_D approach:

The total torque is approximated as:

$$T_c = \sum_{i=1}^z F_c(\theta_i) \frac{D/2}{1000} = z \cdot F_{c,av} \frac{D/2}{1000} \quad [\text{Nm}]$$

Where:

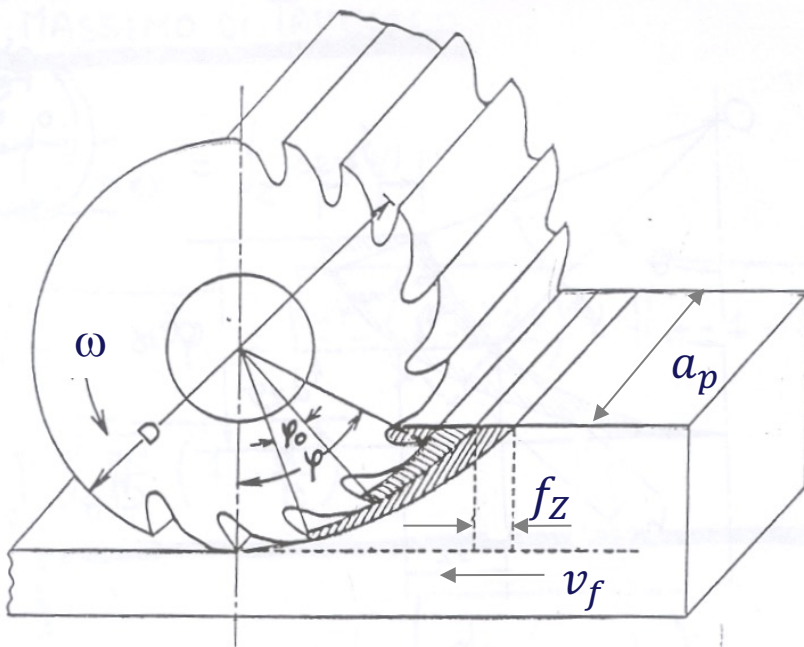
$$F_{c,av} = k_{c,av} \cdot A_{D,av} \quad [\text{N}]$$

$$k_{c,av} = k_{cs} \cdot h_{D,av}^{-x}$$

$$A_{D,av} = h_{D,av} \cdot a_p$$

$$\text{It follows: } P_c = T_c \omega \quad [\text{W}]$$

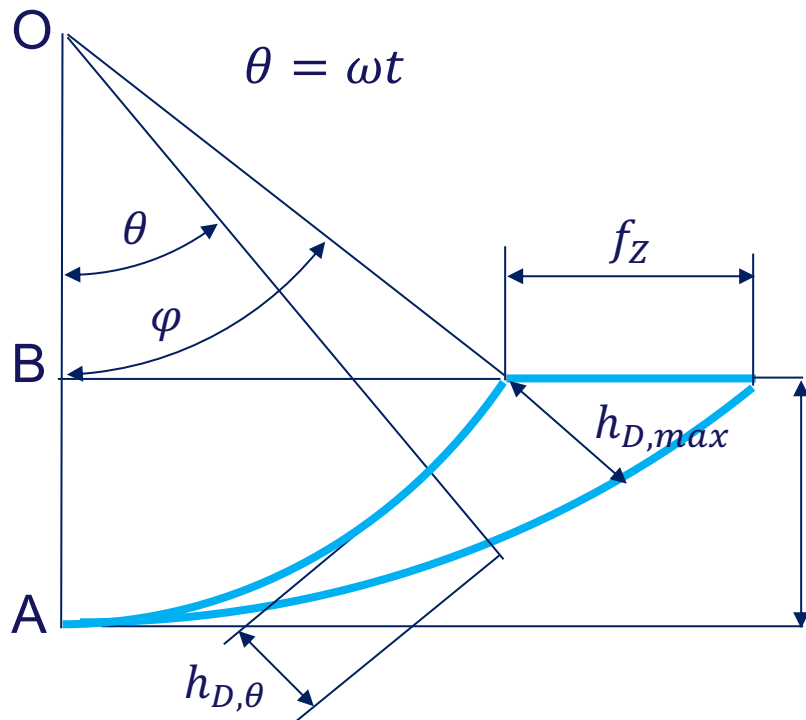
$$\text{Where } \omega = \frac{2\pi n}{60} = \frac{1000 \cdot v_c}{D/2} \quad \left[\frac{\text{rad}}{\text{s}} \right]$$





CUTTING POWER IN SLAB MILLING

If $z > 2$, then it applies the average h_D approach:



$$h_{D,av} = \frac{1}{\varphi} \int_0^{\varphi} h_{D,\theta} d\theta = \frac{1}{\varphi} \int_0^{\varphi} f_z \sin \theta d\theta$$

$$h_{D,av} = \frac{f_z}{\varphi} [-\cos \theta]_0^{\varphi} = \frac{f_z}{\varphi} (1 - \cos \varphi)$$

$$\text{But: } OB = \frac{D}{2} - a_e = \frac{D}{2} \cos \varphi$$

Therefore:

$$(1 - \cos \varphi) = \frac{2a_e}{D}$$

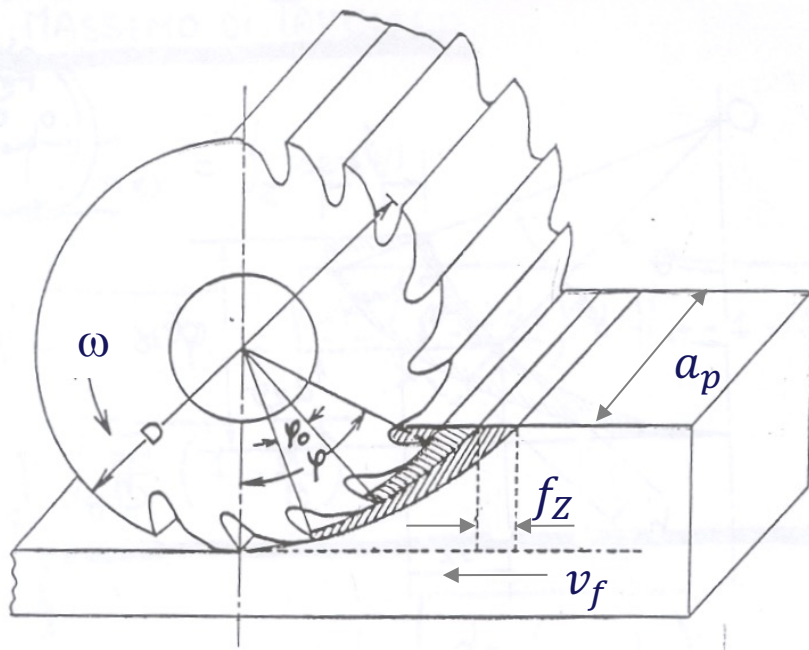
$$h_{D,av} = \frac{f_z}{\varphi} \frac{2a_e}{D}$$

$$A_{D,av} = h_{D,av} \cdot a_p = \frac{f_z}{\varphi} \frac{2a_e a_p}{D}$$



CUTTING POWER IN SLAB MILLING

If $z > 2$, then it applies the average h_D approach:



$$P_c = T_c \omega = z \cdot F_{c,av} \frac{D/2}{1000} \cdot \omega$$

$$P_c = z \cdot k_{c,av} \cdot A_{D,av} \frac{D/2}{1000} \cdot \omega$$

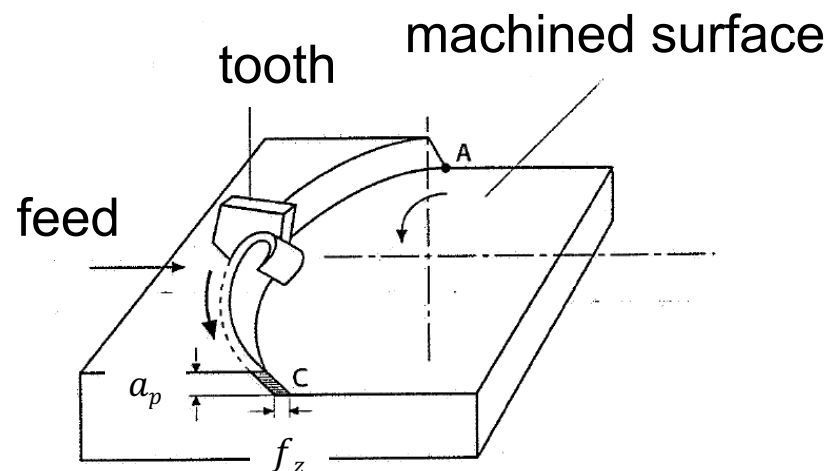
$$P_c = \frac{\varphi}{2\pi} \cdot k_{c,av} \cdot \frac{f_z a_e a_p}{\varphi D/2} \frac{D/2}{1000} \cdot \frac{2\pi n}{60}$$

$$P_c = k_{c,av} \cdot \frac{a_e a_p \cdot Z f_z n}{60 \cdot 1000}$$

$$P_c = k_{c,av} \cdot \frac{a_e a_p \cdot v_f}{60 \cdot 1000} \text{ [W]}$$

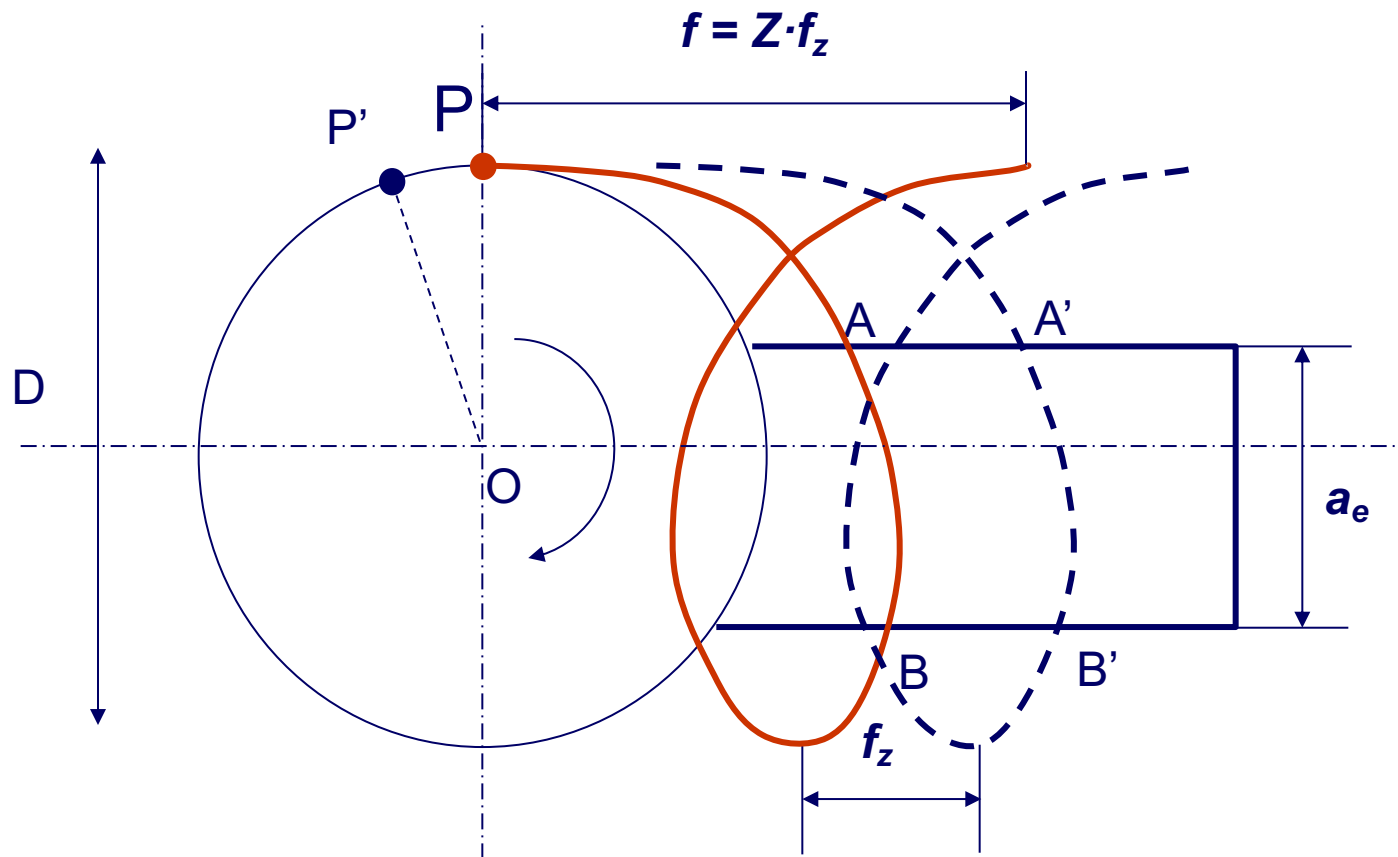


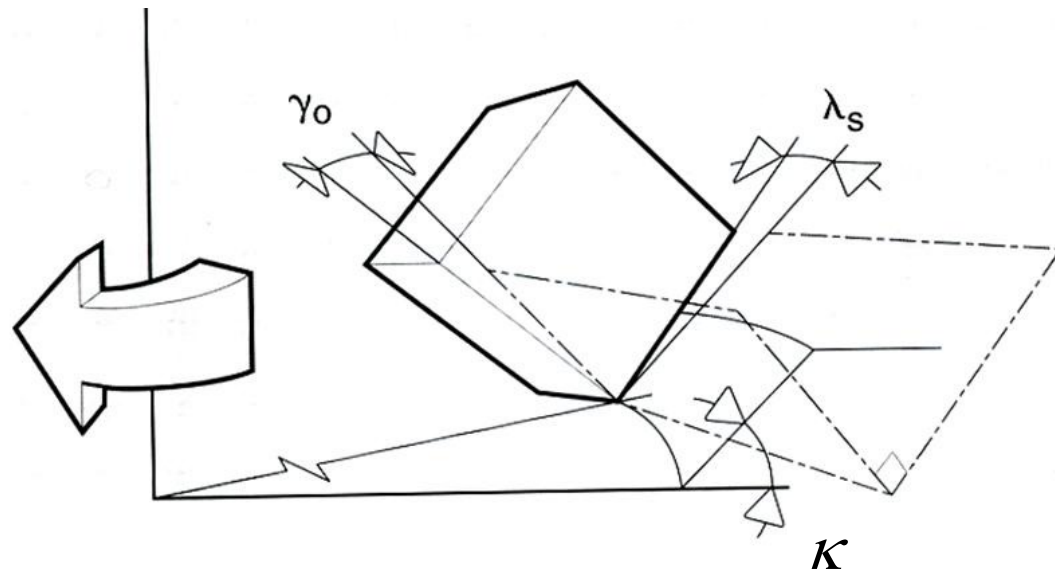
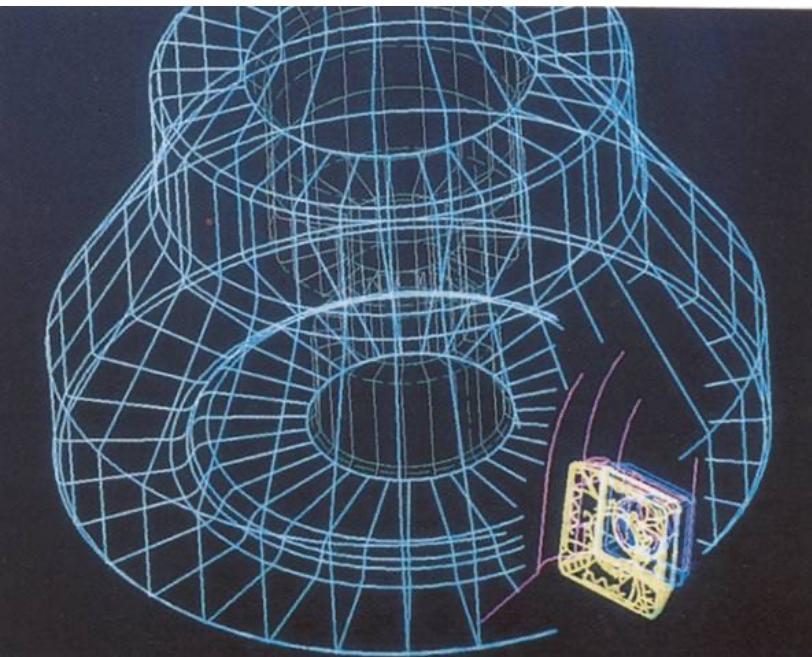
The axis of the mill is perpendicular to the generated surface.
The part of the mill involved in cutting includes also the face part (secondary cutting edges).





The axis of the mill is perpendicular to the generated surface.
The part of the mill involved in cutting includes also the face part (secondary cutting edges).





λ_s back rake angle

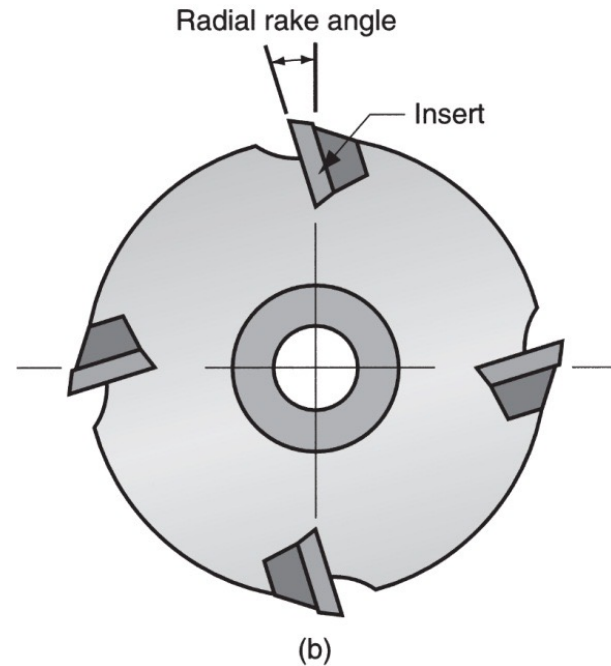
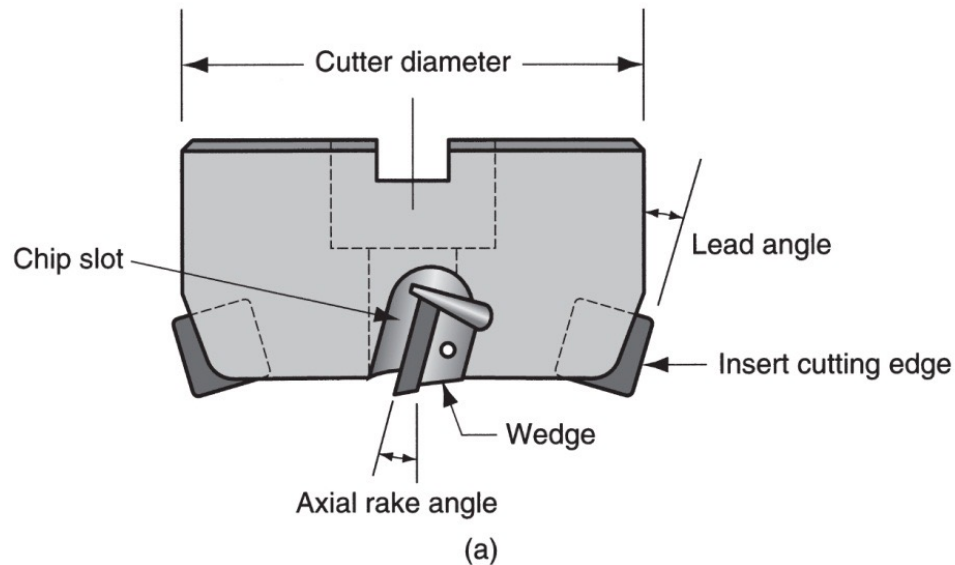
γ_o rake angle

κ entering angle



Tool geometry elements of a four-tooth face milling cutter:

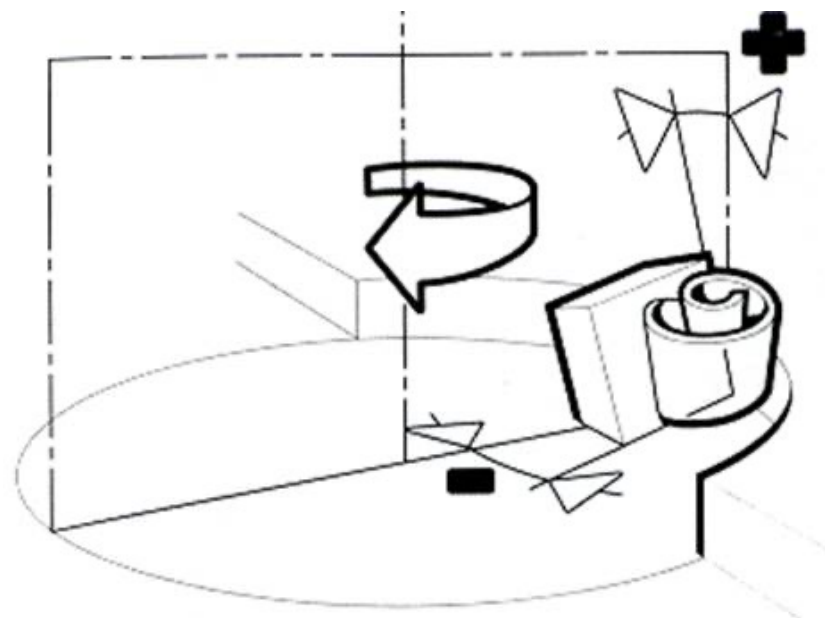
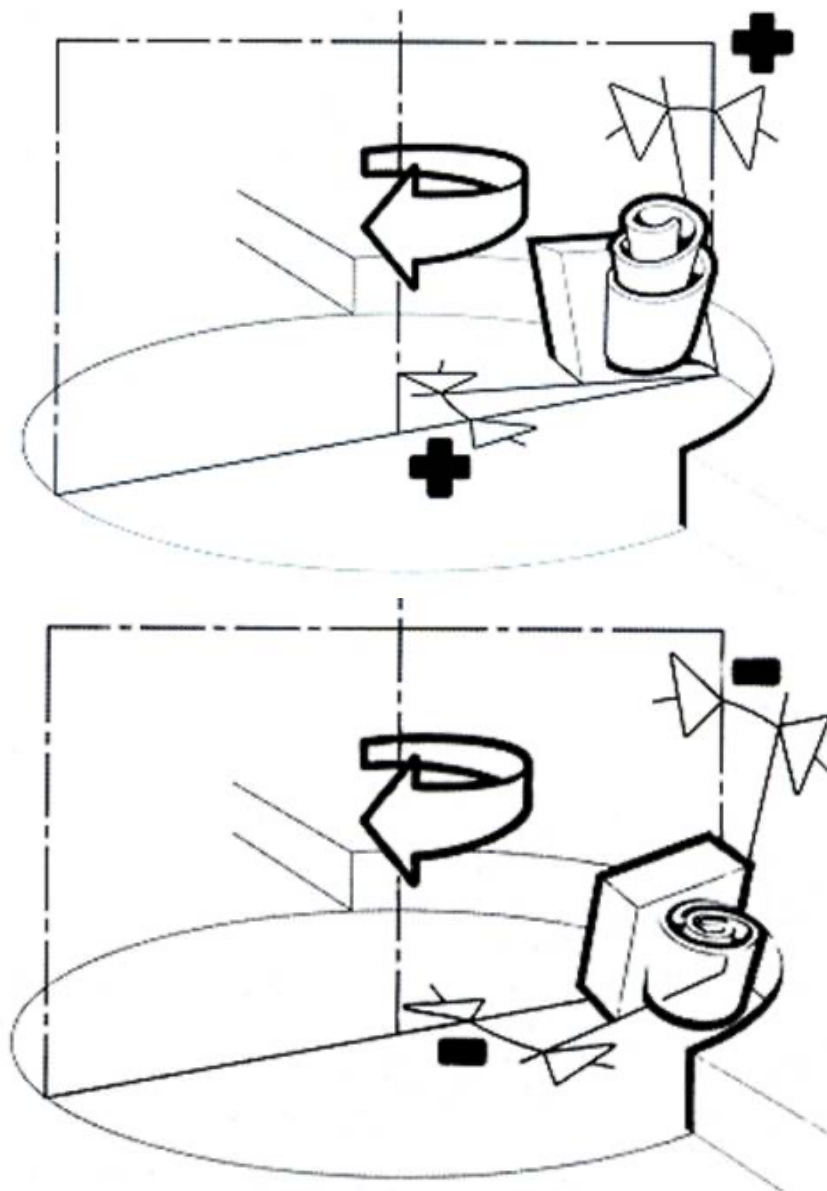
- (a) side view,
- (b) bottom view.





FACE-MILLING CUTTER

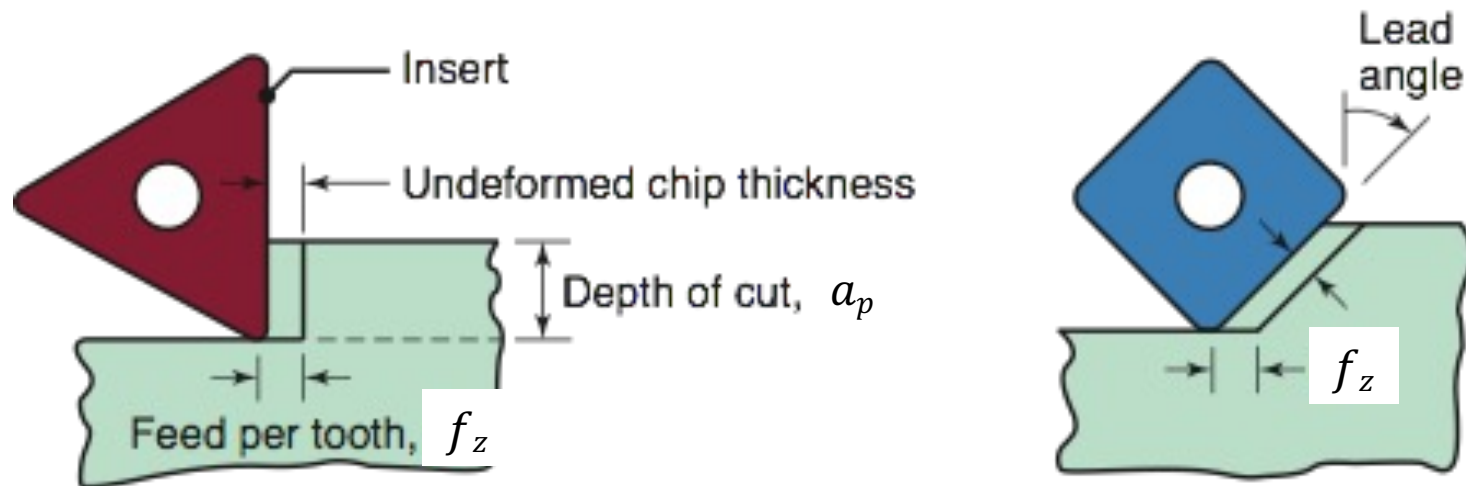
48

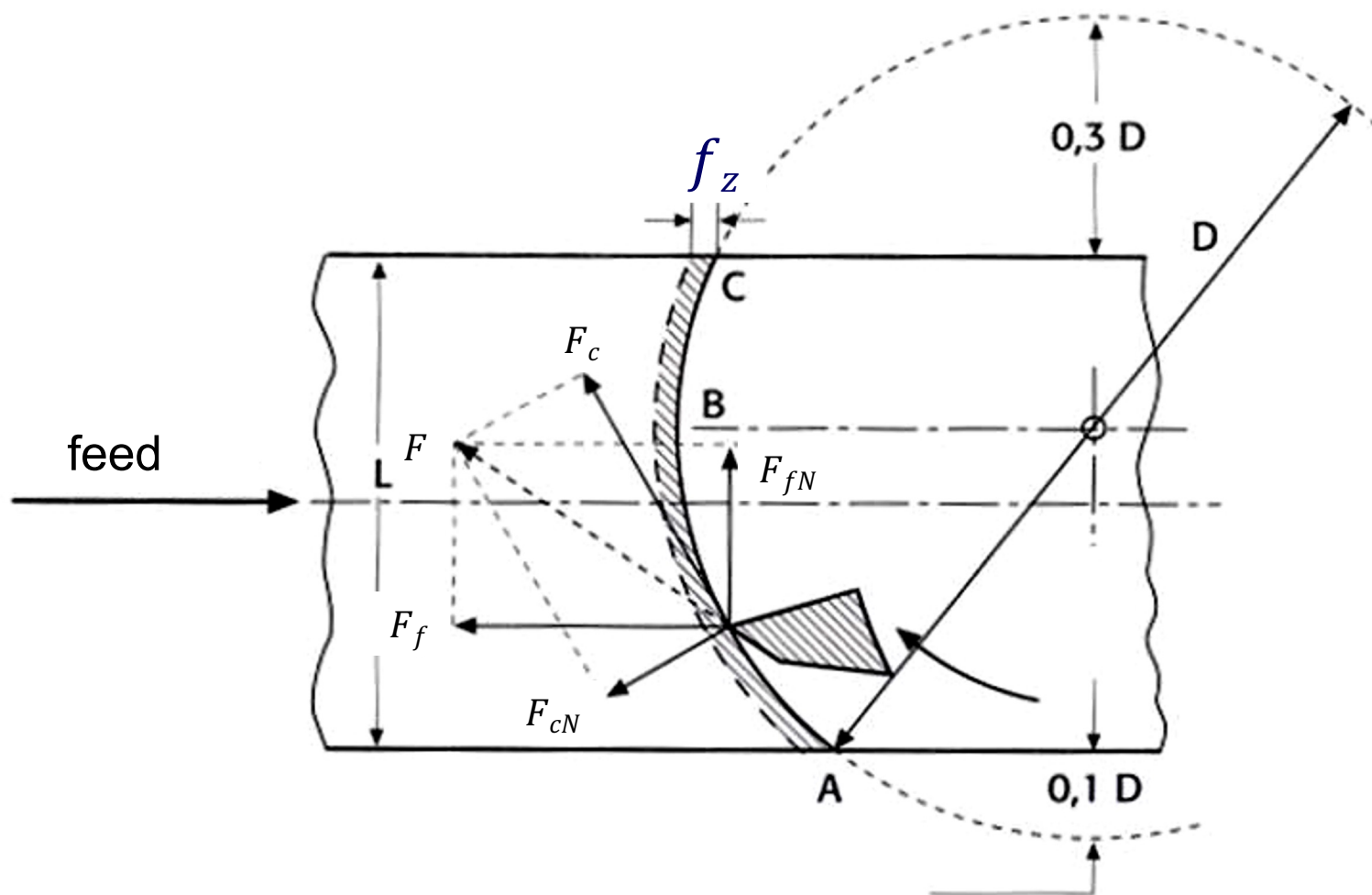


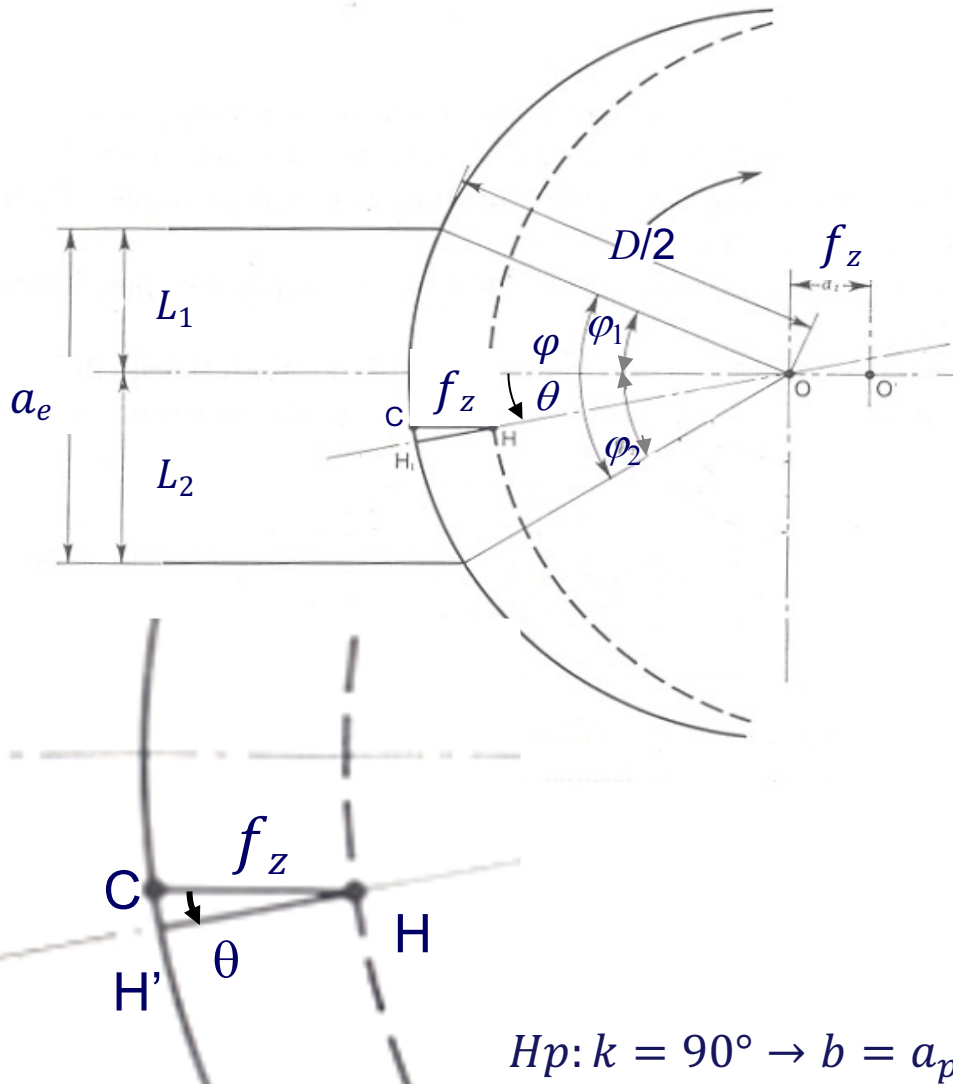


EFFECT OF LEAD ANGLE

The effect of lead angle on the undeformed chip thickness in face milling. As the lead angle increase, the chip thickness decreases, but the length of contact (i.e. chip width) increases.







Chip thickness at time t ,
corresponding to a $\theta = \omega t$
angle, is:

$$h_{D,\theta} = h_D(\theta) = HH' = f_Z \cos \theta$$

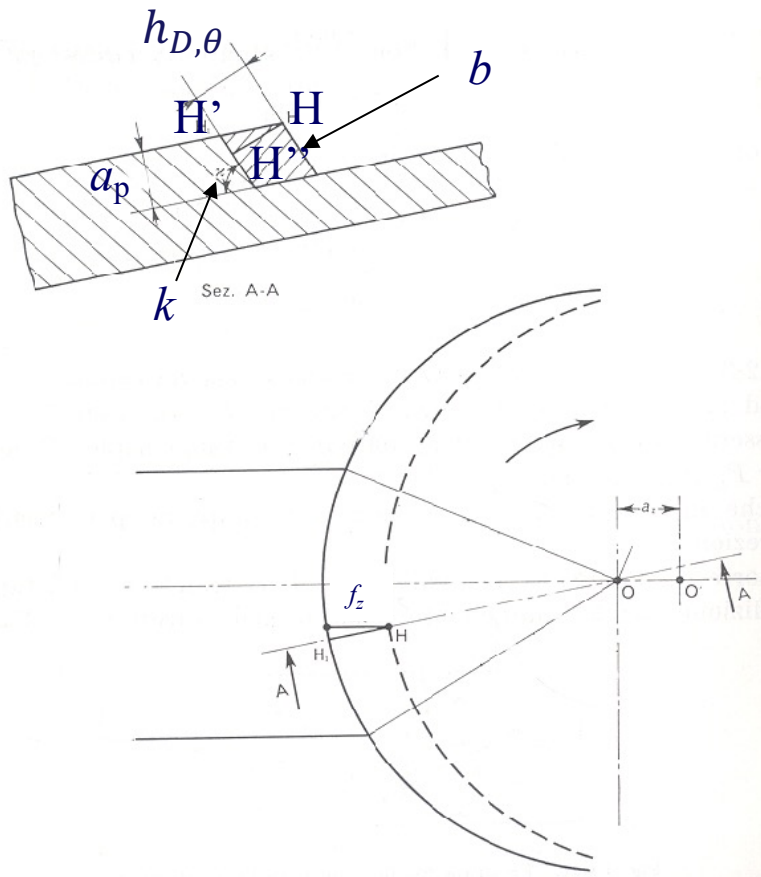
$$h_{D,max} = h_D(\theta = 0) = f_Z$$

Since the chip thickness changes, the chip area (per tooth) changes and, at the time instant t , is:

$$A_{D,\theta} = h_{D,\theta} \cdot a_p$$

$$A_{D,max} = A_D(\theta = 0) = f_Z \cdot a_p$$

$$Hp: k = 90^\circ \rightarrow b = a_p$$



$$H_p: k \neq 90^\circ \rightarrow b = \frac{a_p}{\sin k}$$

$$h_{D,\theta} = h_D(\theta) = HH'' = HH' \cdot \sin k$$

$$h_{D,\theta} = f_Z \cos \theta \cdot \sin k$$

$$h_{D,max} = h_D(\theta = 0) = f_Z \cdot \sin k$$

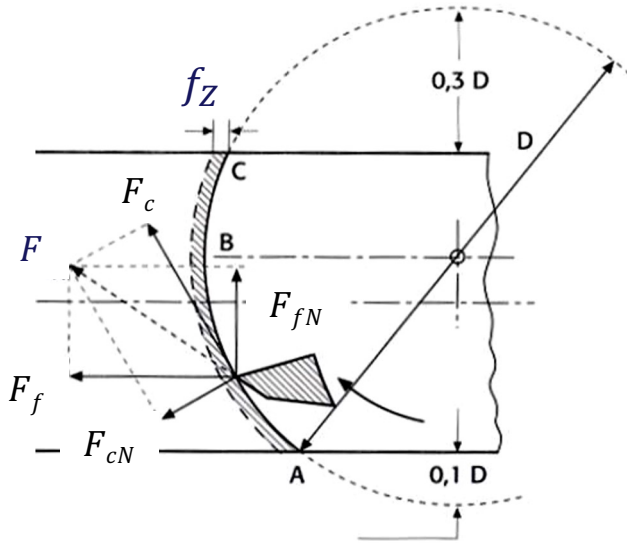
$$A_{D,\theta} = h_{D,\theta} \cdot b = f_Z \cos \theta \cdot \sin k \cdot \frac{a_p}{\sin k}$$

$$A_{D,\theta} = f_Z \cos \theta \cdot a_p$$

$$A_{D,max} = A_D(\theta = 0) = f_Z \cdot a_p$$

NB: A_D does NOT depend on k

Similar to slab milling the cutting force, torque and power in face milling are:



$$F_c(\theta) = k_{c,\theta} A_{D,\theta} \text{ [N]}$$

Where $k_{c,\theta} = k_{cs} h_{D,\theta}^{-x}$

Therefore each tooth gives the following contribution to the torque:

$$T_{c,\theta} = F_c(\theta) \frac{D/2}{1000} = k_{c,\theta} A_{D,\theta} \frac{D/2}{1000} \quad [\text{Nm}]$$

The total torque is:

$$T_c = \sum_{i=1}^Z F_c(\theta_i) \frac{D/2}{1000} \text{ [Nm]}$$

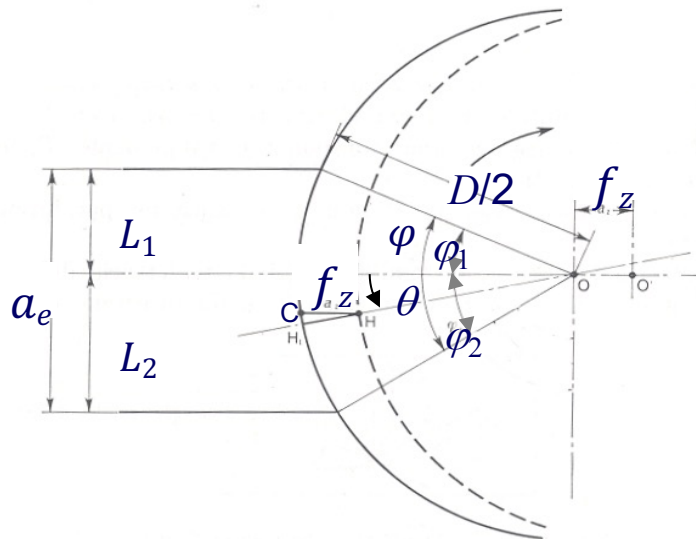
It follows, the cutting power is: $P_c = T_c \omega$ [W]

Where $\omega = \frac{2\pi n}{60} = \frac{1000 \cdot v_c}{D/2} \left[\frac{rad}{s} \right]$



CUTTING POWER IN FACE MILLING

If $z \leq 1$, then it applies the worst case approach



$$h_{D,max} = h_D(\theta = 0) = f_z$$

$$A_{D,max} = A_D(\theta = 0) = f_z \cdot a_p$$

$$F_{c,max} = k_{c,max} A_{D,max} \text{ [N]}$$

$$\text{Where } k_{c,max} = k_{cs} h_{D,max}^{-x}$$

Therefore the torque is:

$$T_c = F_{c,max} \frac{D/2}{1000} = k_{c,max} A_{D,max} \frac{D/2}{1000} \text{ [Nm]}$$

$$\text{Hp: } k = 90^\circ \rightarrow b = a_p$$

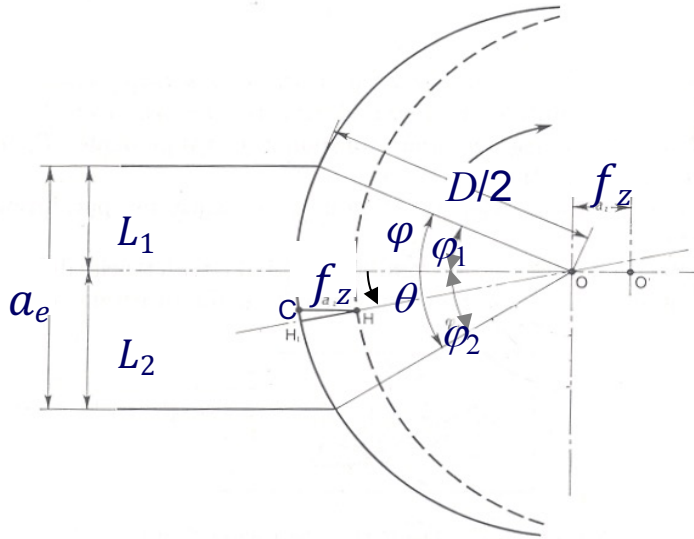
$$\text{It follows: } P_c = T_c \omega \text{ [W]}$$

$$\text{Where } \omega = \frac{2\pi n}{60} = \frac{1000 \cdot v_c}{D/2} \left[\frac{\text{rad}}{\text{s}} \right]$$



CUTTING POWER IN FACE MILLING

If $z > 2$, then it applies the average h_D approach:



Hp: $k = 90^\circ \rightarrow b = a_p$

The total torque is approximated as:

$$T_c = \sum_{i=1}^z F_c(\theta_i) \frac{D/2}{1000} = z \cdot F_{c,av} \frac{D/2}{1000} \quad [\text{Nm}]$$

Where:

$$F_{c,av} = k_{c,av} \cdot A_{D,av} \quad [\text{N}]$$

$$k_{c,av} = k_{cs} \cdot h_{D,av}^{-x}$$

$$A_{D,av} = h_{D,av} \cdot a_p$$

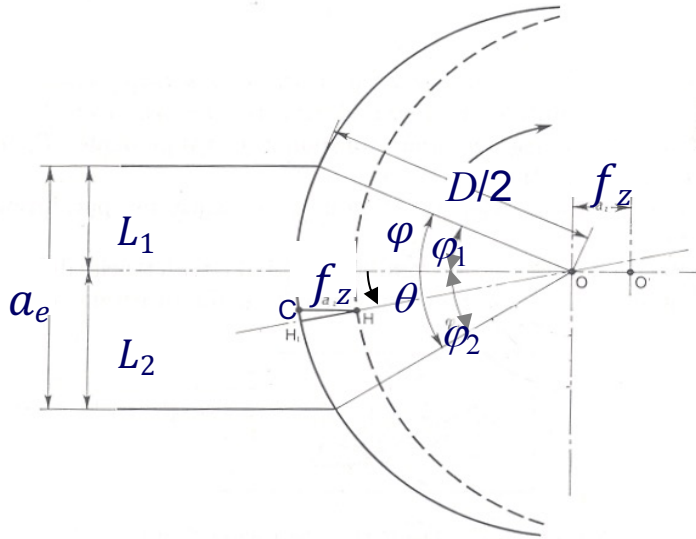
It follows: $P_c = T_c \omega \quad [\text{W}]$

$$\text{Where } \omega = \frac{2\pi n}{60} = \frac{1000 \cdot v_c}{D/2} \quad \left[\frac{\text{rad}}{\text{s}} \right]$$



CUTTING POWER IN FACE MILLING

If $z > 2$, then it applies the average h_D approach:



$$Hp: k = 90^\circ \rightarrow b = a_p$$

$$h_{D,av} = \frac{1}{\phi} \int_{-\phi_1}^{\phi_2} h_{D,\theta} d\theta = \frac{1}{\phi} \int_{-\phi_1}^{\phi_2} f_z \cos \theta d\theta$$

$$h_{D,av} = \frac{f_z}{\phi} [\sin \theta]_{-\phi_1}^{\phi_2} = \frac{f_z}{\phi} (\sin \phi_1 + \sin \phi_2)$$

$$\text{But: } a_e = \frac{D}{2} (\sin \phi_1 + \sin \phi_2)$$

Therefore:

$$(\sin \phi_1 + \sin \phi_2) = \frac{2a_e}{D}$$

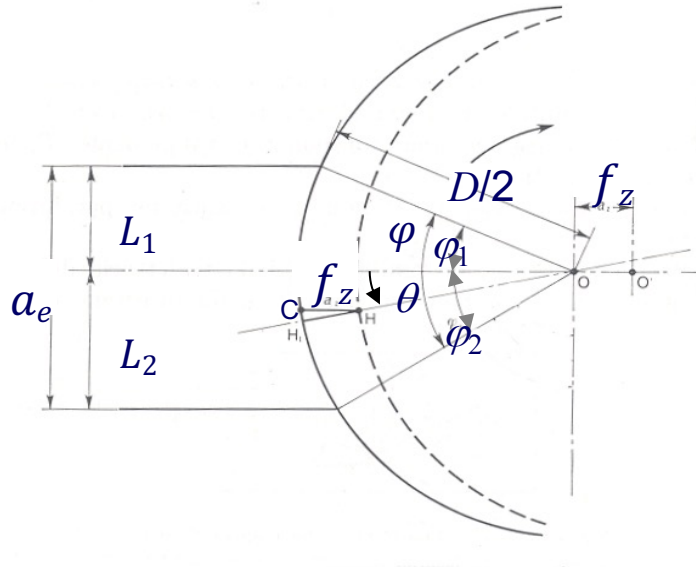
$$h_{D,av} = \frac{f_z}{\phi} \frac{2a_e}{D}$$

$$A_{D,av} = h_{D,av} \cdot a_p = \frac{f_z}{\phi} \frac{2a_e a_p}{D}$$



CUTTING POWER IN SLAB MILLING

If $z > 2$, then it applies the average h_D approach:



$$P_c = T_c \omega = z \cdot F_{c,av} \frac{D/2}{1000} \cdot \omega$$

$$P_c = z \cdot k_{c,av} \cdot A_{D,av} \frac{D/2}{1000} \cdot \omega$$

$$P_c = \frac{\varphi}{2\pi} \cdot k_{c,av} \cdot \frac{f_z a_e a_p}{\varphi} \frac{D/2}{1000} \cdot \frac{2\pi n}{60}$$

$$P_c = k_{c,av} \cdot \frac{a_e a_p \cdot Z f_z n}{60 \cdot 1000}$$

$$P_c = k_{c,av} \cdot \frac{a_e a_p \cdot v_f}{60 \cdot 1000} \text{ [W]}$$

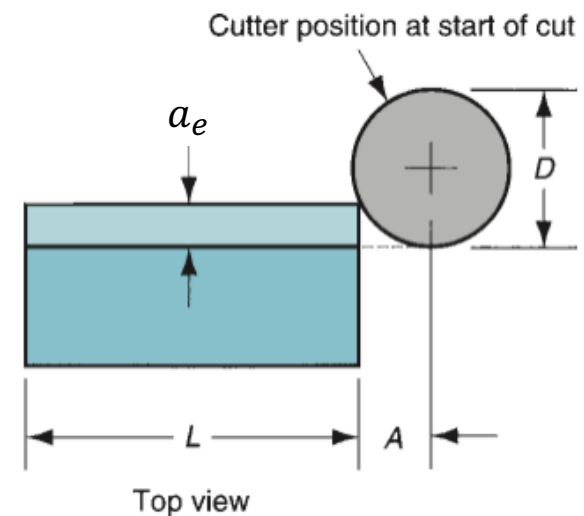
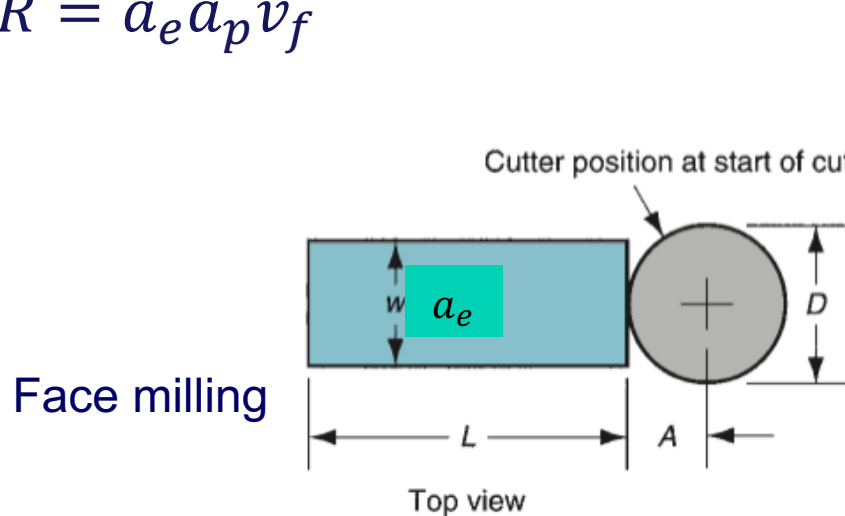
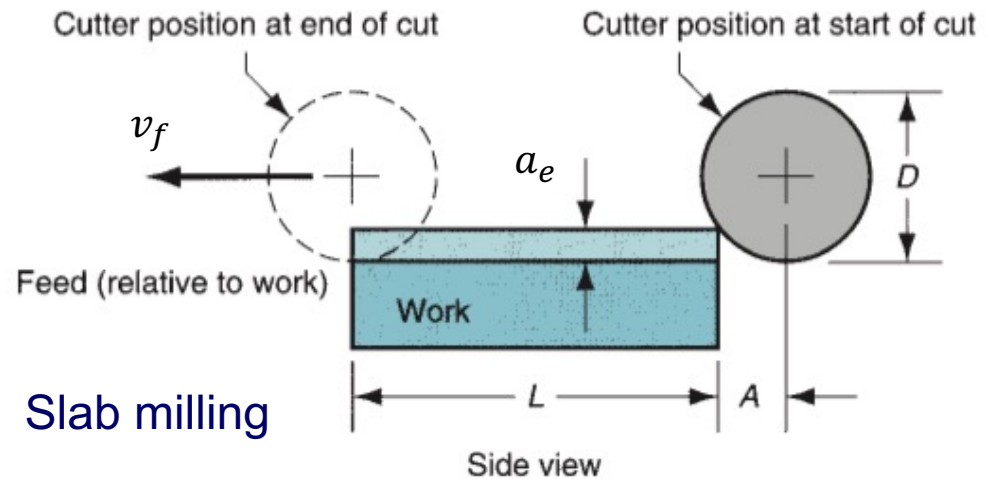


Cutting time:

$$t_c = \frac{L + A}{v_f} = \frac{L + A}{Z f_z n}$$

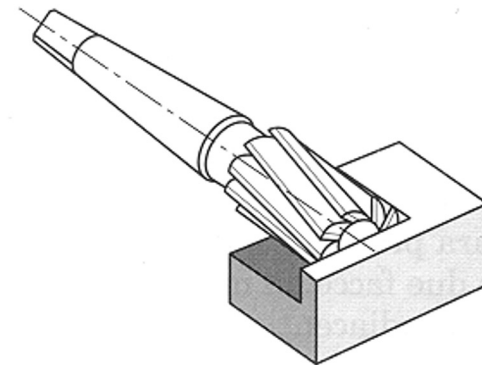
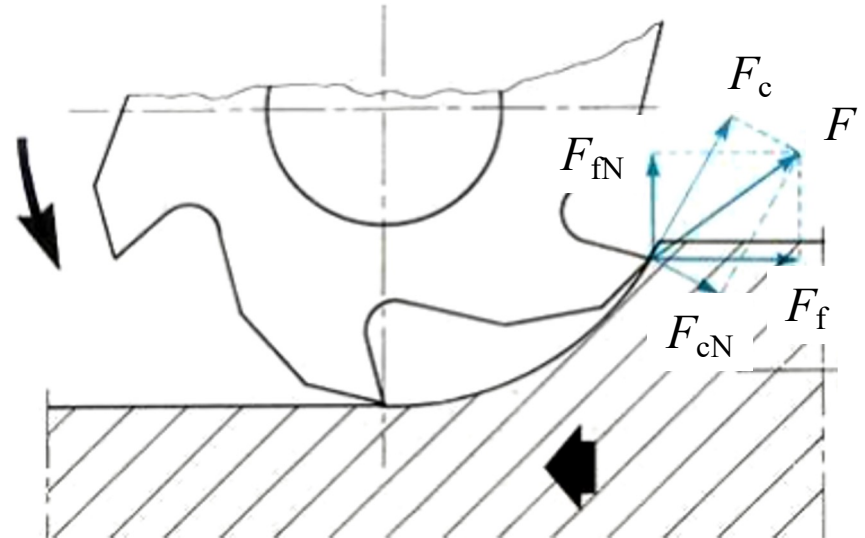
$$t_c = \frac{V}{MRR}$$

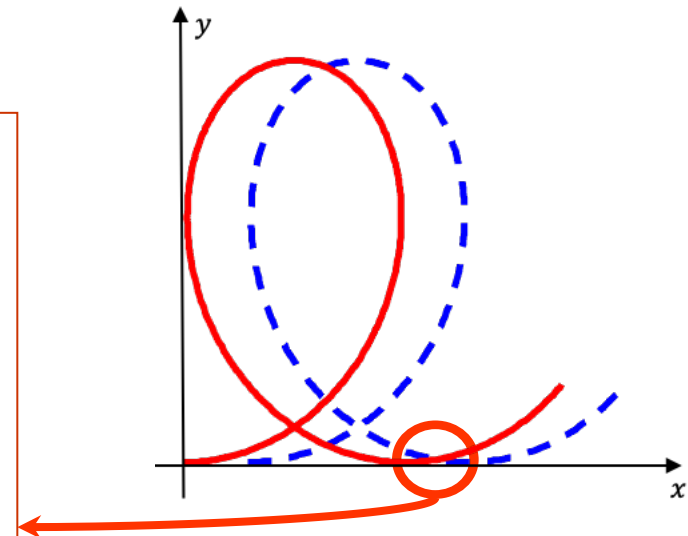
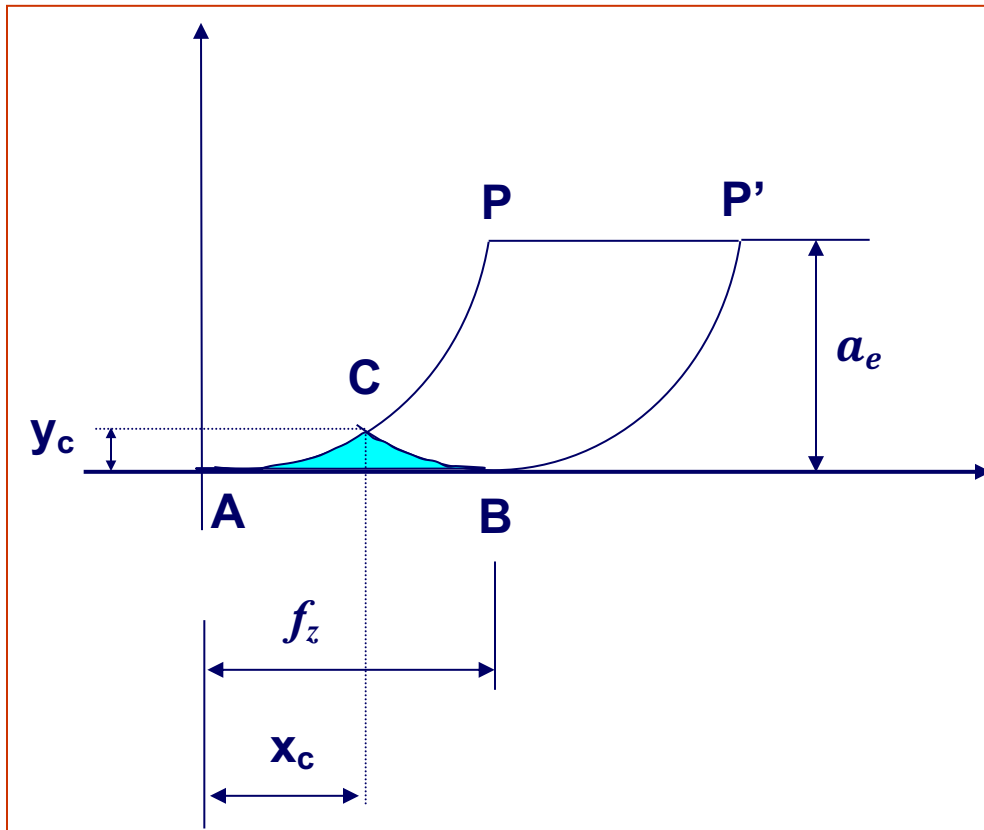
$$MRR = a_e a_p v_f$$





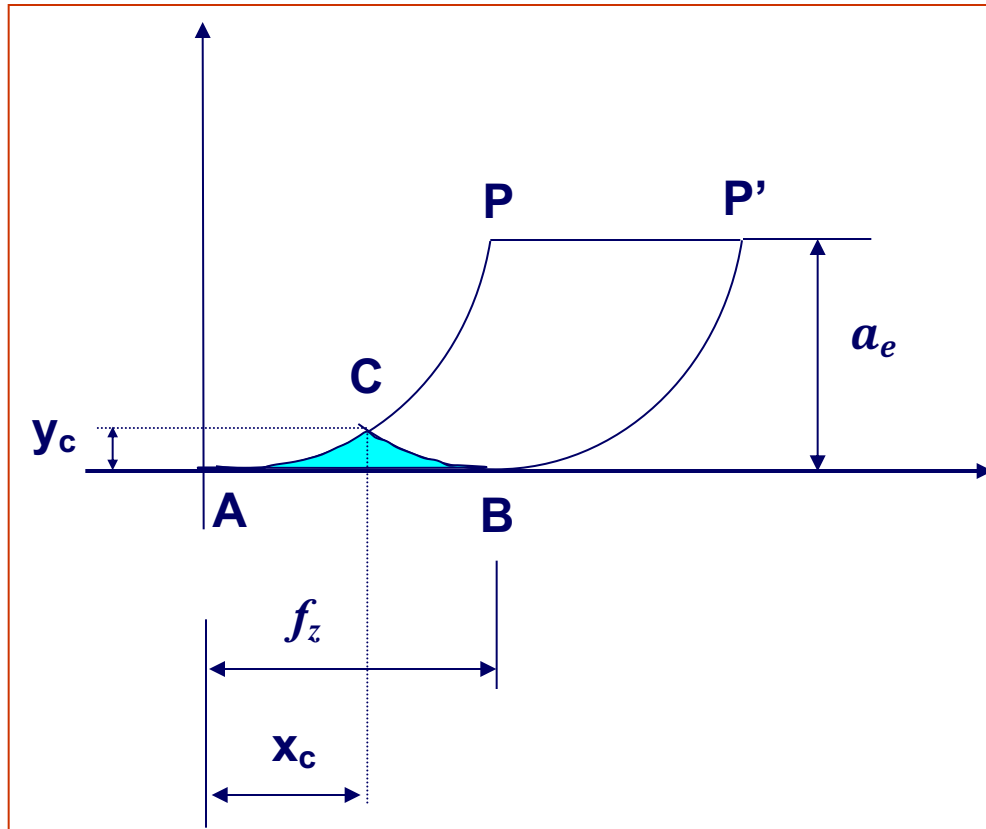
Let's take for example a contour milling with end mill.





Due to symmetry:

$$x_c = \frac{f_z}{2}$$



$$\begin{cases} y = \frac{D}{2}(1 - \cos \omega t) \\ x = \frac{D}{2} \sin \omega t + v_f t \end{cases}$$

It follows that:

$$\frac{f_z}{2} = \frac{D}{2} \sin \omega t_c + v_f t_c$$



Being f_z small $\sin \omega t_c \cong \omega t_c$

$$t_c = \frac{f_z/2}{\omega R + v_f} = \frac{f_z/2}{v_c + v_f}$$

$$\omega t_c = \frac{\omega f_z/2}{\omega R + Z f_z n} = \frac{\omega f_z/2}{\omega R + Z \cdot f_z \frac{60\omega}{2\pi}} = \frac{\pi f_z}{2\pi R + 60 Z f_z}$$

$R_{\max} = y_c$:

$$y_c = R(1 - \cos \omega t_c) = R\left(1 - \sqrt{1 - \sin^2 \omega t_c}\right) \cong R\left(1 - \sqrt{1 - (\omega t_c)^2}\right)$$

$$y_c \cong R\left[1 - \left(1 - \frac{(\omega t_c)^2}{2}\right)\right] = \frac{R}{2}(\omega t_c)^2 = \frac{R}{2}\left(\frac{\pi f_z}{2\pi R + 60 Z f_z}\right)^2$$



SURFACE ROUGHNESS IN SLAB MILLING

$$R_{\max} \cong \frac{R}{2} \frac{\pi^2 f_z^2}{(2\pi R + 60 Z f_z)^2} \cdot 10^3$$

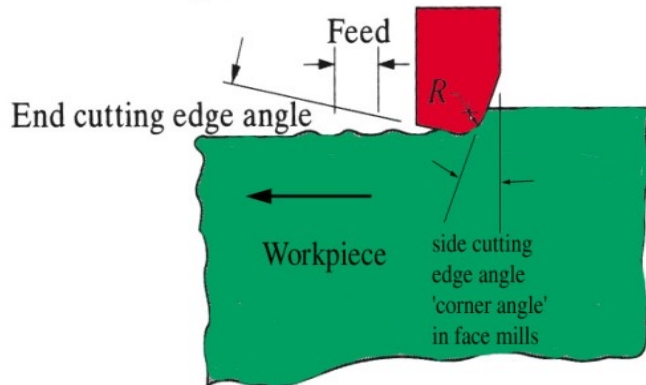
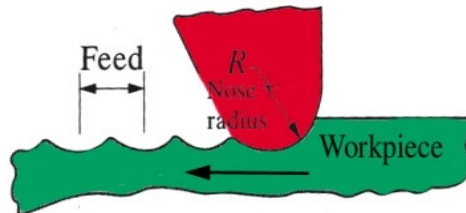
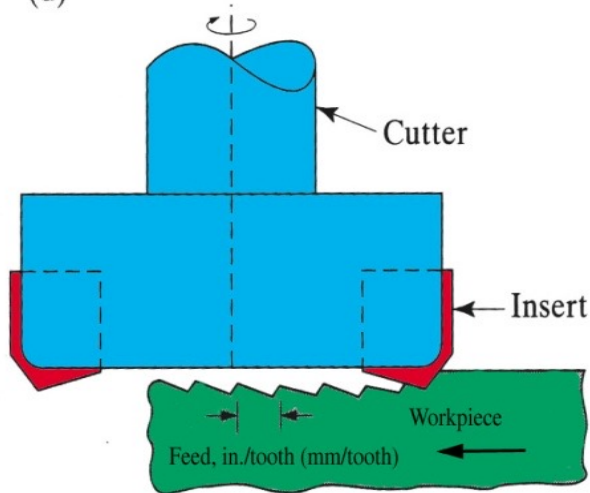
with f_z [mm/rev], R [mm], $R_{\max}$ [μm]

If $2\pi R \gg 60 Z f_z \rightarrow R_{\max} \cong \frac{f_z^2}{8R} \cdot 10^3$

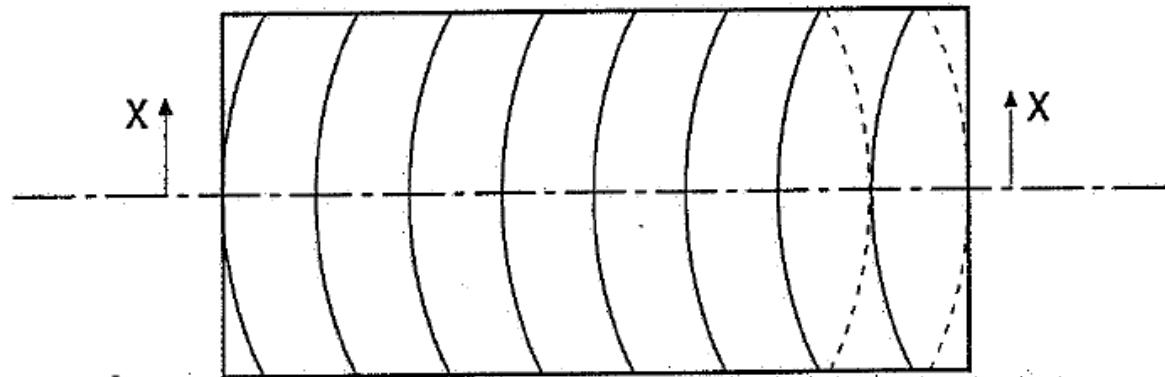
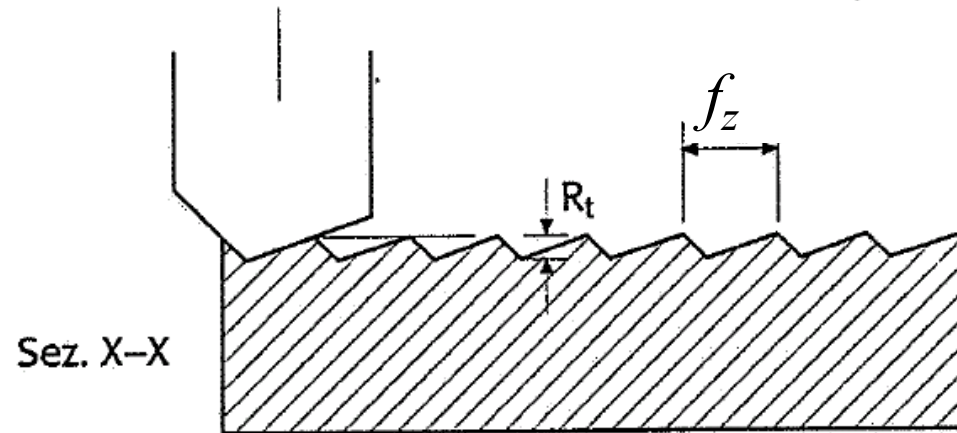
$$R_a \cong \frac{1}{4} R_{\max}$$



(d)

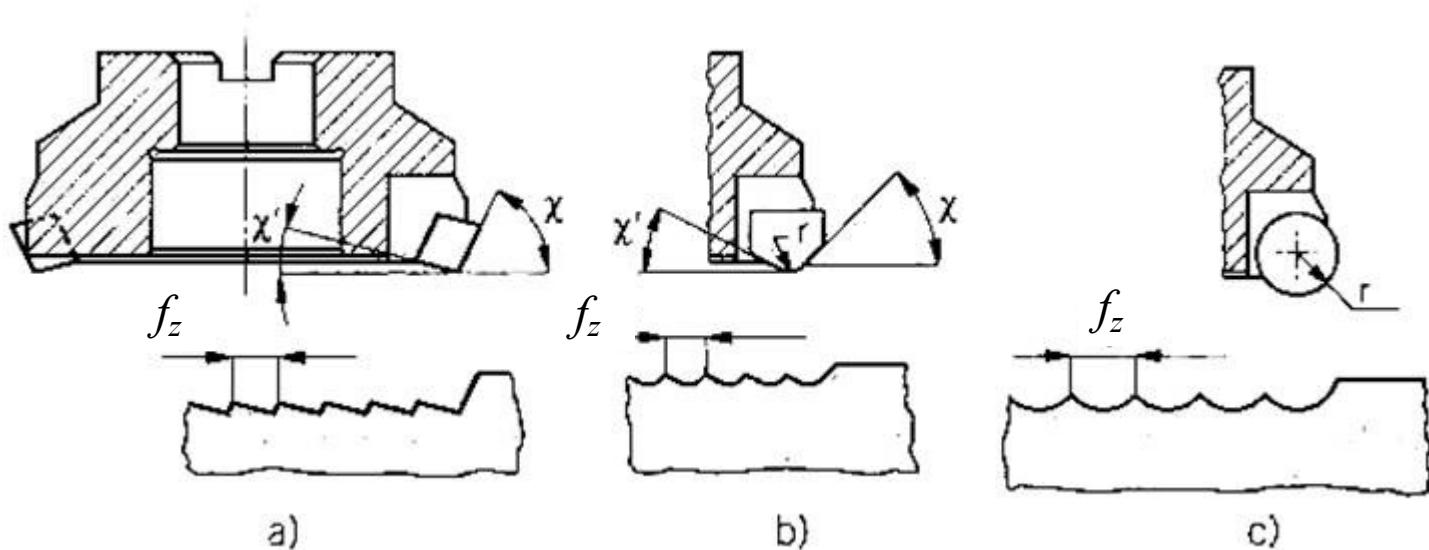


tooth profile





SURFACE ROUGHNESS IN FACE MILLING



$$R_{\max} = \frac{f_z}{\cotg k' + \cotg k} 10^3 \quad [\mu\text{m}]$$

$$R_{\max} = \frac{f_z^2}{8r} 10^3 \quad [\mu\text{m}]$$

with

f_z [mm/(rev*tooth)]

r [mm] tool radius nose